Semilinear spectra and one-type spectra for two-variable logic with Presburger counting

Michael Benedikt, Chia-Hsuan Lu, Tony Tan

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Editorial note. Fixed independently CC BY4.0 full native author version and all actual appendices.

Editorial note. Only original combined Ctwo spectrum theorem counted; all model-construction/core/support proofs retained.

Editorial note. Original unused tkz-berge/tabularray imports omitted from wrapper only. Full original files preserved; no mathematical alteration or local compilation.

Editorial scope. Complete author material collection for one structural theorem; author retains responsibility for the mathematics.

Allowed standard prerequisites (editorial).

- Presburger arithmetic semilinearity and standard two-variable type analysis. Original cited background and exact definitions retained. New differentiated/core partial-model construction and main gluing/model-realization proofs are retained as actual core, not renamed prerequisites.

Abstract

We present new results on finite satisfiability of logics with counting and arithmetic. One result is a tight bound on the complexity of satisfiability of logics with so-called local Presburger quantifiers, which sum over neighbors of a node in a graph. A second contribution concerns computing a semilinear representation of the cardinalities associated with a formula in two variable logic extended with counting quantifiers. Such a representation allows you to get bounds not only on satisfiability for these logics, but for satisfiability in the presence of additional "global cardinality constraints": restrictions on cardinalities of unary formulas, expressed using arbitrary decidability logics over arithmetic. In the process, we provide simpler proofs of some key prior results on finite satisfiability and semi-linearity of the spectrum for these logics.

1 Introduction

This paper concerns the problem of determining whether a formula in some logic $\mathcal{L}$ is satisfied in some finite structure: the *finite satisfiability problem for* $\mathcal{L}$. For the fragments of interest to us, the decidability of this problem will

imply decidability of many other static analysis problems for the logic, such as validity of a sentence over finite structures and equivalence of two sentences over finite structures. The problem is known to be undecidable for first order logic [22]. In the last decades two paradigms for decidability emerged, one based on restricting to *guarded quantification*, and another restricting to *two variable fragments*. The two variable fragment of first order logic was shown to have decidable finite satisfiability problem and this was extended to allow "counting quantifiers", of the form $\exists^{\geq k} y\ \phi(x, y)$, where k is fixed. The complexity of the satisfiability problem for this logic, normally denoted C^2, over finite structures, as well as the complexity of satisfiability when considering arbitrary structures, was isolated in [16]. The same problem for the sublogic where quantification is guarded was shown to have slightly lower complexity [17].

In recent years there has been interest in adding more powerful arithmetic capabilities to these decidable logics. Instead of just asking whether the number of elements satisfying a property is above or below a constant, we can ask whether the cardinality is even, or whether the cardinality of several properties satisfies a linear inequality. *Presburger logic* focuses on linear inequalities between properties, and *local Presburger logic*, GP^2, restricts to properties that involve neighbors of a given element. These logics are most naturally applied to finite models, to avoid dealing with addition or parity of infinity: thus in this work, *finite satisfiability will always be our default*, and we just refer to "satisfiability" henceforward. In [4] it was shown that local Presburger logics are in some sense equally expressive as certain variations of Graph Neural Networks (GNNs), and GNN verification problems can be reduced to satisfiability problems for these logics. It has been shown that satisfiability of Presburger logic is undecidable, while satisfiability of local Presburger logic is decidable [2]: but the exact complexity of local Presburger logic had not been determined. Another way of enhancing logic with arithmetic involves *global cardinality constraints*. In such an enhancement, we can write a constraint that defines a constraint (e.g., a linear inequality) between one variable formulas. But we cannot have such a constraint parameterized by a second free variable. For example, we can write that unary predicate U has cardinality below unary predicate V. But if E and F are two binary predicates, we cannot express that the elements that are not E adjacent to x have greater cardinality than the elements that are not F adjacent to x.

As with local Presburger logic, it follows from prior results that C^2 enhanced with global cardinality constraints has decidable (finite) satisfiability problem. The exact complexity is not explicitly studied in the literature, although in the case of global cardinality constraints consisting of linear inequalities, tight bounds follow from results of Rudolph in [19]. A natural approach to proving decidability for logics enhanced with global constraints follows is to show that the *the spectrum of each tuple of unary formulas of the logic is effectively semilinear*. The spectrum of a tuple of unary formulas $\phi_1(x) \dots \phi_k(x)$ is the set of possible values that the tuple of cardinalities of the satisfiers of the formula can have within a finite model. For example, for the pair of formulas $\phi_1(x) = U(x), \phi_2(x) = U(x) \wedge V(x)$ the spectrum consists of all pairs (m, n) with $n \leq m$. The spectrum being effectively semilinear means that we can find a quantifier-free formula of arithmetic that defines it. From effective semilinearity of the spectrum, we easily obtain decidability with additional global constraints. Here we provide complexity bounds for several of these problems,

one featuring arithmetic over counts of neighbors, another concerning computing the spectrum of formulas. See Table 1 for the summary of our new upper bounds. At the same time, we provide simpler proofs of two fundamental results in the area: decidability of (finite) satisfiability for GP^2 (see Section 4), and effective semi-linearity of the spectrum for C^2 (Section 5). We complement these positive results with a counterexample to semilinearity of spectra for GP^2 unary formulas (Section 6).

2 Preliminaries

Basic notions. A *graph* will always mean a finite directed graph without self-loops. A *colored graph* is a graph where nodes have a unique node label, and every pair of distinct nodes is connected by a unique edge (in one direction or the other), with node and edge vocabularies distinct. A *multigraph* is a finite directed graph without self-loops that allows multiple edges (including no edge) between a pair of distinct nodes. A *colored multigraph* is defined similarly, but requiring each node to have a unique label. The multiplicity of a multigraph is defined as the maximum edge multiplicity within it.

The ith unit vector, denoted by $\mathbf{e}_i$, has i^{th} entry 1, and other entries are 0. We define $\mathbf{0}$ as the vector where all entries are 0, and $\mathbf{1}$ as the vector that all entries are 1. For $\mathbf{v}, \mathbf{u} \in \mathbb{N}^n$, $\mathbf{v} \preccurlyeq \mathbf{u}$ if, for $1 \leq i \leq n$, $\mathbf{v}_i \leq \mathbf{u}_i$ Let $\mathsf{proj}_n(\mathbf{v})$ denote the projection operator that maps $\mathbf{v}$ to its first n entries. Let $\|\mathbf{A}\|$ and $\|\mathbf{v}\|$ denote the maximal absolute values of the entries in $\mathbf{A}$ and $\mathbf{v}$, respectively.

Linear algebra and Kleene star. An *integer linear constraint* is an inequality $\sum_i a_i \cdot x_i \leq c$ involving integer coefficients a_i, non-negative integer variables x_i, and an integer constant c. The set of solutions of such a constraint is called a *modulus-free semilinear set*, while if we include equations $x = c \mod p$ for integers $c, p > 0$ it is a *semilinear set*. A constraint is *homogeneous* if $c = 0$.

For an integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$, we denote by $[\![\mathcal{Q}(\mathbf{x})]\!]$ the set of solutions in $\mathbb{N}$, i.e., $[\![\mathcal{Q}(\mathbf{x})]\!] := \{\mathbf{v} \in \mathbb{N}^n | \ \mathbf{A}\mathbf{v} = \mathbf{c}\}$, and similarly for a Boolean combination of linear systems. Note that if $\mathcal{Q}(\mathbf{x})$ is feasible, then it has a solution in which the number of non-zero assignments is bounded, and the values of the assignments are also bounded.

Lemma 1 ([13], [6], also see Corollary 2.2 in [12])**.** *There are constants $c_1, c_2 \in \mathbb{N}$ such that for every integer linear system $\mathcal{Q}(\mathbf{x})$, if $\mathcal{Q}(\mathbf{x})$ admits a solution in $\mathbb{N}$, then it admits a solution in $\mathbb{N}$ in which the number of variables assigned non-zero values is at most $c_1 t \log(c_2 t M)$ and every variable is assigned a value bounded by $c_1 t (tM)^{c_2 t}$, where t is the number of constraints in $\mathcal{Q}(\mathbf{x})$ and M is the maximal absolute value of the coefficients in $\mathcal{Q}(\mathbf{x})$.*

Definition 2 (Kleene star)**.** *For a set $\mathcal{S} \subseteq \mathbb{N}^n$, the* Kleene star *of $\mathcal{S}$ is defined as*

$$\mathsf{KleeneStar}(\mathcal{S}) \ := \ \left\{ \sum_{s \in \mathcal{S}'} s \ \middle| \ \mathcal{S}' \text{ is a finite multisubset of } \mathcal{S} \right\}.$$

Note that $\mathbf{0} \in \mathsf{KleeneStar}(\mathcal{S})$ for any $\mathcal{S}$.

If $\mathcal{S}$ is semilinear then its Kleene star is semilinear: this is closely-related to Parikh's theorem which gives a correspondence between semi-linear sets and

regular languages: see [14, 8]. We will be interested in bounding the complexity of the Kleene star of a semilinear set using numerical data associated to the set. Note that for an integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$, if $\mathbf{c} = \mathbf{0}$, then $\mathsf{KleeneStar}([\![\mathcal{Q}(\mathbf{x})]\!]) = [\![\mathcal{Q}(\mathbf{x})]\!]$. The following result gives a bound when $\mathbf{c} \neq \mathbf{0}$.

Lemma 3. *For every integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$ with $\mathbf{c} \neq \mathbf{0}$, where $\mathbf{A} \in \mathbb{Z}^{m\times n}$ and $\mathbf{c} \in \mathbb{Z}^m$, there exists $\tilde{\mathbf{A}} \in \mathbb{Z}^{(n+t)\times(n+k)}$ with $\left\|\tilde{\mathbf{A}}\right\| = 1$ such that $\mathsf{KleeneStar}([\![\mathcal{Q}(\mathbf{x})]\!]) = \left[\!\left[\tilde{\mathcal{Q}}(\mathbf{x})\right]\!\right]$, where $\tilde{\mathcal{Q}}(\mathbf{x})$ is the Boolean combination of integer linear systems*

$$\exists \mathbf{y}_1, \mathbf{y}_2 \, (\mathbf{x} = \mathsf{proj}_n(\mathbf{y}_1) + \mathbf{y}_2) \wedge (\tilde{\mathbf{A}}\mathbf{y}_1 = \mathbf{0}) \wedge (\mathbf{A}\mathbf{y}_2 = \mathbf{0}) \wedge ((\mathbf{y}_1 = \mathbf{0}) \rightarrow (\mathbf{y}_2 = \mathbf{0})),$$

$k = n \cdot (2D+1)^m + 1$, $t \leq 2nk$, and $D = n \cdot \|\mathbf{A}\| \cdot ((n+1) \cdot \|\mathbf{A}\| + \|\mathbf{c}\| + 1)^m$. Moreover, $\tilde{\mathbf{A}}$ can be computed in time $2^{\mathcal{O}\left(\log n + \log K + m^2\right)}$, where $K := \max(\|\mathbf{A}\|, \|\mathbf{c}\|)$.

We defer the proof of this to the appendix.

Types. Let σ be a vocabulary with unary $U_1, \ldots, U_n$ and binary $R_1, \ldots, R_m$.

Definition 4 (1- and 2-type)**.** *A* 1-type *(w.r.t. σ) is a maximal consistent subset of*

$$\bigcup_{i\in[n]} \{U_i(x), \neg U_i(x)\} \cup \bigcup_{i\in[m]} \{R_i(x,x), \neg R_i(x,x)\}.$$

A 2-type *(w.r.t. σ) is a maximal consistent subset of*

$$\bigcup_{i\in[m]} \{R_i(x,y), \neg R_i(x,y), R_i(y,x), \neg R_i(y,x)\}.$$

We denote the set of 1-types by OneTps and the set of 2-types TwoTps.

Given an element v in a σ-structure $\mathcal{G}$, we let $\mathsf{OneTp}(v)$ denote its one-type in $\mathcal{G}$ and similarly for a pair of elements v, u in $\mathcal{G}$ we let $\mathsf{TwoTp}(v,u)$ denote its two-type in $\mathcal{G}$; in both cases we omit the dependence on $\mathcal{G}$ for brevity.

The following definitions are variations of notions in Pratt-Hartmann's [16, 17, 18].

Definition 5 (Silent 2-type)**.** *The* silent type*, denoted by η_0, is the unique 2-type that consists of only negated terms. If a type is not silent we say it is* audible*.*

Definition 6 (Dual of 2-type)**.** *For a 2-type η, the* dual *of η, denoted by $\widetilde{\eta}$, it the unique 2-type that swaps x and y in η.*

We denote the set of all 2-types except η_0 by TwoTps^+, and the set of all 2-types that contain $R_t(x,y)$ by TwoTps_t for $1 \leq t \leq m$.

Note that $|\mathsf{OneTps}| = 2^{n+m}$, $|\mathsf{TwoTps}| = 2^{2m}$, $\left|\mathsf{TwoTps}^+\right| = 2^{2m} - 1$, and $|\mathsf{TwoTps}_t| = 2^{2m-1}$.

We can abstract a model based on types:

Definition 7 (Type graph)**.** *We associate to a σ-structure in a binary vocabulary a colored graph in an exponentially larger vocabulary: each vertex is colored with a 1-type, and each edge is colored with a 2-type. This is the* type graph *of the σ-structure.*

Losing information, we can abstract a finite model based on cardinalities of 1-types. The notion will be crucial in Section 5:

Definition 8 (1-type cardinality vector of a σ-structure)**.** *Let $\mathcal{G}$ be a σ-structure. The 1-type cardinality vector of $\mathcal{G}$ is a $|\mathsf{OneTps}|$-dimensional vector where the ith component is the number of elements in $\mathcal{G}$ with 1-type π_i.*

3 Logics and prior complexity bounds concerning them

We fix a vocabulary σ which consists of n unary predicates $U_1, \ldots, U_n$ and m binary predicates $R_1, \ldots, R_m$. All of our logics will consist of formulas with at most two variables, x and y. We start by reviewing *two variable logic with counting* (C^2), built up from atomic formulas $U_i(x)$, $U_i(y)$, $R_i(x,y)$, and $R_i(y,x)$ via the Boolean operators and the quantifier constructs $\rho(x) := \exists^{\geq k} y\ \phi(x,y)$, where ϕ is a C^2 formula and k is a natural number, and similarly with the role of x and y swapped.

Our semantics will always be over *finite structures* interpreting the vocabulary σ. The formula $\rho(y)$ above holds in a model M at an element u for x when there are least k elements v such that $M \models \phi(u,v)$. The case $k = 1$ represents standard existential quantification.

Guarded two variable logic with counting (GC^2), restricts C^2 by requiring quantification to be guarded by an atom. The quantifier rule now includes the forms:

$$\exists^{\geq k} y\ R_i(x,y) \wedge \phi(x,y), \quad \exists^{\geq k} y\ R_i(y,x) \wedge \phi(x,y), \quad \text{or} \quad \exists^{\geq k} x\ \ U_i(x) \wedge \phi(x).$$

The logic P^2 replaces C^2's counting quantifier with a *Presburger quantifier*:

$$\rho(x) \ := \ \left(\sum_{t \in [m]} \kappa_t \cdot \#_y \left[\phi_t(x,y)\right] \right) \ \circledast \ \delta,$$

where κ_t are integers, $\circledast$ is an inequality or equality, and δ is an integer. In P^2 we will also allow *unguarded quantification over unary formulas*: if $\phi(x)$ is a formula, so is $\exists x\ \phi(x)$. For example, $\exists x \left(\#_y \left[R(x,y)\right] - 3 \cdot \#_y \left[R(y,x)\right] = 0\right)$ is a sentence of P^2 that holds in a directed graph when there is an element that has 3 times as many outgoing edges as incoming edges.

The logic GP^2 (*guarded two variable logic with Presburger quantifiers*) restricts P^2 by requiring each ϕ_i in a Presburger quantifier to be guarded, although still allowing unguarded unary quantification. Note that global cardinality constraints can be expressed in P^2, but not in GP^2. Following prior papers [12], we do not allow a quantifier that tests the modulus of a linear combination of cardinalities of one dimensional sets, and thus our quantifiers are in a sense weaker than Presburger arithmetic. A *Presburger modulus constraint* is as above, but allowing $\circledast\ \delta$ to be replaced by $(= m \mod n)$, for natural numbers m, n with $m < n$. Likewise, C^{2+g} can be extended to allow modulus constraints. In the appendix we show that *all of of our results on GP^2 and C^{2+g} also hold for the extension with modulus constraints*; in the body of the paper we focus on plain GP^2 and C^{2+g} in the statements.

Given a unary formula $\phi(x)$ in some logic, and a finite model M, we let $|\phi|_M$ be the cardinality of ϕ in M; we extend this notation to a tuple $\langle \phi_1(x) \dots \phi_k(x) \rangle$. Given such a tuple, its *spectrum* is the set of vectors of numbers of the form $|\phi_1(x) \dots \phi_k(x)|_M$. We say that such a spectrum is *semi-linear* if it can be represented as a set of linear inequalities. When we talk about *computing the spectrum* for a formula, we mean computing a semi-linear representation. If we have such a representation, we can clearly check whether an individual $\phi_i(x)$ is satisfiable, simply by seeing if the corresponding set of constraints is feasible. Thus, in cases, where the spectrum is known always to have such a representation, computing the spectrum can be seen as a generalization of the satisfiability pboelm.

The positive results in this paper, as well as the prior bounds, are summarized in Table 1. In the table, Sat Logic refers to the satisfiability problem for formulas of the logic over finite structures, while Spectrum refers to a bound on the size of a representation of the spectrum, again over finite structure. An entry of the form X-c indicates that the problem is complete for complexity class X.

	Sat Logic	Spectrum Size Bound
GC^2	EXP-c [17]	EXP (formerly 2EXP)
C^2	NEXP-c [16]	EXP, **Thm 25** (formerly 2EXP [3])
GP^2	**EXP-c, Thm 9** (formerly EXP-hard, in 3NEXP [2])	Open
P^2	Undecidable [2]	No size bound

Table 1: Complexity results, with those new to this work in bold.

We give results on the spectrum, but they have applications to decidability of richer logics. C^2 can be extended with *global unary cardinality constraints*: these are of the form $\rho := \left(\sum_{t\in[n]} \kappa_t \cdot |\phi_t(x)|\right) \circledast \delta$, where ϕ_t are C^2 formulas in one free variable, κ_t are integers, $\circledast$ is an inequality or equality, and δ is an integer. ρ is a sentence, and we define the semantics only for finite structures M. Such M satisfies ρ if, letting s_t be the number of elements in M satisfying $\phi_t(x)$, we have $\left(\sum_{t\in[n]} \kappa_t \cdot s_t\right) \circledast \delta$. We write C^{2+g} for the extension of C^2 with global cardinality constraints. As we explain later, computation of the spectrum provides an approach to get tight bounds on these extended logics. In the case where the global constraints are in Presburger arithmetic, these can be seen as an alternative to the technique of [19], which obtains tight bounds for the logic through reduction to satisfiability of C^2 itself.

4 Finite satisfiability of GP^2 is in EXP

The goal of this section is to prove:

Theorem 9. *The finite satisfiability problem of* GP^2 *is in* EXP.

GP^2 infrastructure. We fix a vocabulary σ which consists of n unary predicates $U_1, \dots, U_n$ and m binary predicates $R_1, \dots, R_m$. We consider GP^2 sentences φ over σ in *normal form*:

$$\varphi := \forall x\, \gamma(x) \wedge \bigwedge_{i\in[m]} \forall x \forall y\ (R_i(x,y) \wedge x \neq y) \rightarrow \alpha_i(x,y) \wedge \bigwedge_{i\in[n]} \forall x\, U_i(x) \rightarrow P_i(x),$$

where $\gamma(x)$ is a quantifier-free formula, each $\alpha_i(x,y)$ is a quantifier-free formula, and each $P_i(x)$ is a Presburger quantified formula of the form:

$$P_i(x) \;:=\; \left(\sum_{t\in[m]} \lambda_{i,t} \cdot \#_y \left[R_t(x,y) \wedge x \neq y \right] \right) \;\circledast_i\; \delta_i.$$

Note that in the normal form we allow $\forall x \forall y \;\; (R(x,y) \wedge x \neq y) \rightarrow \alpha(x,y)$ which is equivalent to a GP^2 sentence.

There is a linear time algorithm that converts every GP^2 sentence into an equi-satisfiable sentence in normal form [12], which is based on the standard renaming technique for GC^2 introduced in [10, 17]. See the appendix for an argument.

Definition 10 (Compatibility of 1- and 2-types)**.** *For a GP^2 sentence φ in the normal form above and a 1-type π, we say π is* compatible *with φ, if $\pi(x) \models \gamma(x)$, where $\gamma(x)$ is from the normal form, that is, the "unary universal part" of φ.*

For 1-types π_1 and π_2, and a 2-type η, we say that $\langle \pi_1, \eta, \pi_2 \rangle$ is compatible *with φ, if*

- $\pi_1(x) \wedge \eta(x,y) \wedge \pi_2(y) \models \bigwedge_{i\in[m]} (R_i(x,y) \wedge x \neq y) \rightarrow \alpha_i(x,y)$, *and*
- $\pi_2(x) \wedge \widetilde{\eta}(x,y) \wedge \pi_1(y) \models \bigwedge_{i\in[m]} (R_i(x,y) \wedge x \neq y) \rightarrow \alpha_i(x,y)$.

That is, the tuple satisfies the "binary universal part".

We denote the set of 1-types compatible with φ by $\mathsf{OneTps}^{\varphi} \subseteq \mathsf{OneTps}$. Note that it takes time $\mathcal{O}(|\varphi|)$ to check compatibility with φ either for a 1-type π or a tuple $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps} \times \mathsf{TwoTps} \times \mathsf{OneTps}$.

In order to capture the counting conditions, we introduce *behavior vectors*. Informally, the behavior vector of a vertex encodes the configuration of its neighbors. We fix an ordering of the set $\mathsf{TwoTps}^{+} \times \mathsf{OneTps}$, which has cardinality $2^{n+3m} - 2^{n+m}$. A behavior vector $\mathbf{f}$ is an element of $\mathbb{N}^{2^{n+3m}-2^{n+m}}$. Abusing notation, we write $\mathbf{f}(\pi,\eta)$ to refer to the ith component of $\mathbf{f}$, where i is the index of the tuple $\langle \pi, \eta \rangle$ in the fixed ordering of $\mathsf{TwoTps}^{+} \times \mathsf{OneTps}$.

Definition 11 (Behavior vector of a vertex)**.** *For a σ-structure $\mathcal{G}$ and a vertex v in $\mathcal{G}$, the behavior vector realized by v, denoted by $\mathsf{bh}(v)$, is an element of $\mathbb{N}^{2^{n+3m}-2^{n+m}}$ defined as follows: for every $\eta \in \mathsf{TwoTps}^{+}$ and $\pi \in \mathsf{OneTps}$, $\mathsf{bh}(v)(\eta,\pi)$ is the number of edges e in $\mathcal{G}$ adjacent to v such that the 2-type of e is η and the 1-type of e's destination is π.*

Definition 12 (Characteristic system)**.** *For GP^2 sentence φ in normal form and 1-type π, the characteristic system of φ and π, denoted $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$, is the integer linear system defined:*

- *The variables of $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$ are $x_{\eta,\pi'}$ for every $\eta \in \mathsf{TwoTps}^{+}$ and $\pi' \in \mathsf{OneTps}$.*
- *(Compatibility) For every $\eta \in \mathsf{TwoTps}^{+}$ and $\pi' \in \mathsf{OneTps}$ if π' or $\langle \pi, \eta, \pi' \rangle$ is not compatible with φ, then $x_{\eta,\pi'} = 0$ is a constraint in $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$.*
- *(Counting) For $1 \leq i \leq n$, if $U_i(x) \in \pi$ then the following constraint is in $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$:*

$$\left(\sum_{t\in[m]} \lambda_{i,t} \cdot \sum_{\eta\in\mathsf{TwoTps}_t} \sum_{\pi'\in\mathsf{OneTps}} x_{\eta,\pi'} \right) \;\circledast_i\; \delta_i.$$

Note that there are $\left|\mathsf{TwoTps}^{+} \times \mathsf{OneTps}\right|$ variables and at most n, the number of unary predicates, constraints in $\mathcal{C}_\pi^\varphi(\mathbf{x})$. The absolute value of coefficients in $\mathcal{C}_\pi^\varphi(\mathbf{x})$ is bounded by K, where K is the maximal absolute value of coefficients in φ.

Definition 13 (Compatibility of behavior vector)**.** *For a* GP^2 *sentence* φ *in normal form and a 1-type* π*, we say that a behavior vector* $\mathbf{f}$ *is* compatible *with* φ *and* π *if* $\mathbf{f}$ *is a solution of* $\mathcal{C}_\pi^\varphi(\mathbf{x})$*.*

We denote the set of all behavior vectors that are compatible with φ and π by $\mathcal{B}_\pi^\varphi := [\![\mathcal{C}_\pi^\varphi(\mathbf{x})]\!]$. Note that $\mathcal{B}_\pi^\varphi$ can be infinite but semilinear.

The following property restates satisfaction of a formula in a σ-structure in terms of the type graph of the structure, where the types are identified with vertex- and edge-labels:

Proposition 14. *For every* GP^2 *sentence* φ *in normal form,* φ *is finitely satisfiable if and only if there exists a colored graph* $\mathcal{G}$ *satisfying the following conditions:*

- *(1-type) For every vertex* $v \in V$*,* $\mathsf{OneTp}(v)$ *is compatible with* φ*.*
- *(2-type) For every pair of vertices* $v_1, v_2 \in V$*,* $\langle \mathsf{OneTp}(v_1), \mathsf{TwoTp}(v_1, v_2), \mathsf{OneTp}(v_2)\rangle$ *is compatible with* φ*.*
- *(Counting) For every vertex* $v \in V$*,* $\mathsf{bh}(v)$ *is compatible with* $\mathsf{OneTp}(v)$ *and* φ*.*

We say that a colored multiple graph in the vocabulary of types, $\mathcal{G}$, is a *finite pseudo-model* of a GP^2 sentence φ if it satisfies all conditions in Proposition 14. Here we show that to establish the finite satisfiability of φ, it is sufficient to construct a pseudo-model.

Lemma 15. *For every* GP^2 *sentence* φ *in normal form,* φ *is finitely satisfiable if and only if it has a finite pseudo-model.*

Proof. Note that the type graph of a finite model of φ is a finite pseudo-model of φ. Thus we focus on the if direction, constructing (the type graph of) a finite model of φ using a *duplicating and swapping* procedure.

Let $\mathcal{G}_0$ be a finite pseudo-model of φ, and let $\mathcal{G}_0'$ be a copy of $\mathcal{G}_0$. Define $\mathcal{G}_1$ as the disjoint union of $\mathcal{G}_0$ and $\mathcal{G}_0'$, after iteratively applying the following swapping procedure: for every pair of vertices v and u in $\mathcal{G}_0$ with more than one edge between them, let e be an edge between them, and let e' be the corresponding edge in $\mathcal{G}_0'$. We then swap the destinations of e and e' in $\mathcal{G}_1$.

It is straightforward to verify that $\mathcal{G}_1$ is a pseudo-model of φ and its multiplicity is one less than $\mathcal{G}_0$. We can repeat this procedure until we obtain a finite pseudo-model $\mathcal{G}_t$ where no two vertices have multiple edges. Finally, the type graph of the finite model of φ is obtained by connecting all remaining pairs in $\mathcal{G}_t$ with the silent type. □

The main construction. For every GP^2 sentence φ in normal form, we assume that for every 1-type $\pi \in \mathsf{OneTps}^\varphi$, $\mathbf{0} \notin \mathcal{B}_\pi^\varphi$. Otherwise φ is trivially satisfiable by a model with only one vertex with 1-type π.

Lemma 16. *A* GP^2 *sentence* φ *in normal form is finitely satisfiable if and only if there exists vectors* $\mathbf{g}_\pi \in \mathsf{KleeneStar}(\mathcal{B}_\pi^\varphi)$ *for each* $\pi \in \mathsf{OneTps}^\varphi$ *satisfying the following conditions.*

- *(Matching) For every* $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps}^{\varphi} \times \mathsf{TwoTps}^{+} \times \mathsf{OneTps}^{\varphi}$, $\mathbf{g}_{\pi_1}(\eta, \pi_2) = \mathbf{g}_{\pi_2}(\widetilde{\eta}, \pi_1)$.
- *(Non-triviality) There exists some* $\pi \in \mathsf{OneTps}^{\varphi}$ *such that* $\mathbf{g}_{\pi} \neq \mathbf{0}$.

Proof. **Suppose that φ is finitely satisfiable with finite model $\mathcal{G}$.** For a 1-type π, let $V_\pi \subseteq V$ be the set of vertices v in $\mathcal{G}$ with 1-type π. Note that the V_π partition V. Since $\mathcal{G} \models \varphi$, for every v in V_π, $\mathsf{bh}(v) \in \mathcal{B}^{\varphi}_{\pi}$. We claim that $\mathbf{g}_\pi = \sum_{v \in V_\pi} \mathsf{bh}(v) \in \mathsf{KleeneStar}(\mathcal{B}^{\varphi}_{\pi})$ satisfies the conditions.

- (Matching) For every $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps}^{\varphi} \times \mathsf{TwoTps}^{+} \times \mathsf{OneTps}^{\varphi}$, note that

$$\begin{aligned}\sum_{v \in V_{\pi_1}} \mathsf{bh}(v)(\eta, \pi_2) &= \sum_{v \in V_{\pi_1}} |\{u \in V |\ v \neq u,\ \mathsf{TwoTp}(v,u) = \eta, \text{ and } \mathsf{OneTp}(u) = \pi_2\}| \\ &= \left|\left\{(v,u) \in V^2 \middle|\ v \neq u,\ \mathsf{OneTp}(v) = \pi_2,\ \mathsf{TwoTp}(v,u) = \eta, \text{ and } \mathsf{OneTp}(u) = \pi_2\right\}\right|.\end{aligned}$$

  That is, $\mathbf{g}_{\pi_1}(\eta, \pi_2)$ is the number of edges with types $\langle \pi_1, \eta, \pi_2 \rangle$ in $\mathcal{G}$. Similarly, $\mathbf{g}_{\pi_2}(\widetilde{\eta}, \pi_1)$ is the number of edges with types $\langle \pi_2, \widetilde{\eta}, \pi_1 \rangle$ in $\mathcal{G}$. Since the number of edges with types $\langle \pi_1, \eta, \pi_2 \rangle$ and $\langle \pi_2, \widetilde{\eta}, \pi_1 \rangle$ should be the same, the Matching constraint holds.
- (Non-triviality) Since $\mathcal{G}$ is non-empty, by our assumption, it has a vertex v with 1-type π satisfying that $\mathsf{bh}(v) \neq \mathbf{0}$. Therefore, $\mathbf{g}_\pi = \mathsf{bh}(v) + \sum_{u \in V_\pi \setminus \{v\}} \mathsf{bh}(u) \neq \mathbf{0}$.

Suppose that there exists such $\mathbf{g}_\pi \in \mathsf{KleeneStar}(\mathcal{B}^{\varphi}_{\pi})$ for $\pi \in \mathsf{OneTps}^{\varphi}$. By the definition of Kleene star, for every $\pi \in \mathsf{OneTps}^{\varphi}$, there exists $\mathbf{f}_{\pi,1}, \ldots, \mathbf{f}_{\pi,\ell_\pi} \in \mathcal{B}^{\varphi}_{\pi}$, not necessarily distinct, such that $\mathbf{g}_\pi = \sum_{i \in [\ell_\pi]} \mathbf{f}_{\pi,i}$. We construct a colored multigraph $\mathcal{G}$, which will be the type graph of our structure, as follows. The vertices in $\mathcal{G}$ are $\bigcup_{\pi \in \mathsf{OneTps}^{\varphi}} \{v_{\pi,1}, \ldots, v_{\pi,\ell_\pi}\}$. Each vertex $v_{\pi,i}$ is colored with a 1-type π, and we let V_π be the set of vertices colored with π. For every $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps}^{\varphi} \times \mathsf{TwoTps}^{+} \times \mathsf{OneTps}^{\varphi}$, we add edges between V_{π_1} and V_{π_2} such that for each $v_{\pi_1,i}$, the number of edges connected to it with 2-type η is $\mathbf{f}_{\pi_1,i}(\eta, \pi_2)$ and for each $v_{\pi_2,i}$, the number of edges connected to it with 2-type η is $\mathbf{f}_{\pi_2,i}(\widetilde{\eta}, \pi_1)$. Note that by the Matching constraint, $\mathbf{g}_{\pi_1}(\eta, \pi_2) = \mathbf{g}_{\pi_2}(\widetilde{\eta}, \pi_1)$, so this construction is always possible. We claim that $\mathcal{G}$ is a pseudo-model of φ. Note that each vertex in $\mathcal{G}$ is colored with 1-type in $\mathsf{OneTps}^{\varphi}$, each edge in $\mathcal{G}$ is colored with 2-type in TwoTps^{+}, and it is routine to check that the behavior vector of $v_{\pi,i}$ is $\mathbf{f}_{\pi,i}$. By the Non-triviality constraint, $\mathcal{G}$ is non-empty. Thus by Lemma 15, φ is finitely satisfiable. □

Above we connected the finite satisfiability problem of GP^2 sentence φ and the Kleene star of behavior vectors. However, because the cardinality of the set $\mathsf{KleeneStar}(\mathcal{B}^{\varphi}_{\pi})$ is *infinite*, it is not clear how to determine the solvability of the conditions in Lemma 16. In addition, this representation does not tell us anything about the shape of the models.

Recall that the set of behavior vectors $\mathcal{B}^{\varphi}_{\pi}$ is the solution set of the characteristic system $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$. We can apply Lemma 3 and obtain that

$$\mathsf{KleeneStar}(\mathcal{B}^{\varphi}_{\pi}) = \mathsf{KleeneStar}(\llbracket \mathcal{C}^{\varphi}_{\pi}(\mathbf{x}) \rrbracket) = \left\llbracket \tilde{\mathcal{C}}^{\varphi}_{\pi}(\mathbf{x}) \right\rrbracket.$$

Thus the conditions in Lemma 16 can be encoded in the following integer linear system.

Definition 17. *For every* GP^2 *sentence* φ *in normal form, the integer linear system* $\mathcal{Q}^\varphi(\mathbf{x})$ *is defined as follows.*

- *The variables of* $\mathcal{Q}^\varphi(\mathbf{x})$ *are* x_{π_1,η,π_2} *for every* $\pi_1,\pi_2 \in \mathsf{OneTps}^\varphi$ *and* $\eta \in \mathsf{TwoTps}^+$*. Let* $\mathbf{x}_{\pi_1}$ *denote the vector of variables* x_{π_1,η,π_2}*.*
- *(Structure) For every* $\pi \in \mathsf{OneTps}^\varphi$*,* $\tilde{\mathcal{C}}^\varphi_\pi(\mathbf{x}_\pi)$ *is part of* $\mathcal{Q}^\varphi(\mathbf{x})$*, where* $\tilde{\mathcal{C}}^\varphi_\pi(\mathbf{x})$ *is the integer linear system obtained by applying Lemma 3 to the characteristic system* $\mathcal{C}^\varphi_\pi(\mathbf{x})$*.*
- *(Matching) For every* $\langle \pi_1,\eta,\pi_2\rangle \in \mathsf{OneTps}^\varphi \times \mathsf{TwoTps}^+ \times \mathsf{OneTps}^\varphi$*, the following constraint is in* $\mathcal{Q}^\varphi(\mathbf{x})$*,* $x_{\pi_1,\eta,\pi_2} = x_{\pi_2,\widetilde{\eta},\pi_1}$*, where we recall that* $\widetilde{\eta}$ *denotes the dual 2-type.*
- *(Non-triviality)* $\sum_{\pi_1\in\mathsf{OneTps}^\varphi}\sum_{\eta\in\mathsf{TwoTps}^+}\sum_{\pi_2\in\mathsf{OneTps}^\varphi} x_{\pi_1,\eta,\pi_2} > 0$*.*

Lemma 18. *For every* GP^2 *sentence* φ *in normal form,* φ *is finitely satisfiable if and only if* $\mathcal{Q}^\varphi(\mathbf{x})$ *has a solution in* $\mathbb{N}$*.*

Proof. **Suppose that φ is finitely satisfiable**. By Lemma 16, there exists $\mathbf{g}_\pi \in \mathsf{KleeneStar}(\mathcal{B}^\varphi_\pi)$ for $\pi \in \mathsf{OneTps}^\varphi$ satisfying the Matching and Non-triviality conditions.

For every $\pi_1,\pi_2 \in \mathsf{OneTps}^\varphi$ and $\eta \in \mathsf{TwoTps}^+$, let $a_{\pi_1,\eta,\pi_2} := \mathbf{g}_{\pi_1}(\eta,\pi_2)$. We claim that assigning x_{π_1,η,π_2} to a_{π_1,η,π_2} gives a solution of the system $\mathcal{Q}^\varphi(\mathbf{x})$.

- (Structure) By definition, for every $\pi \in \mathsf{OneTps}^\varphi$, $\mathbf{g}_\pi \in \mathsf{KleeneStar}(\mathcal{B}^\varphi_\pi)$. By Lemma 3, $\mathsf{KleeneStar}(\mathcal{B}^\varphi_\pi) = \mathsf{KleeneStar}([\![\mathcal{C}^\varphi_\pi(\mathbf{x})]\!]) = \left[\!\!\left[\tilde{\mathcal{C}}^\varphi_\pi(\mathbf{x})\right]\!\!\right]$. Thus, $\mathbf{a}_\pi$ is a solution of $\tilde{\mathcal{C}}^\varphi_\pi(\mathbf{x})$.
- (Matching) For every $\langle \pi_1,\eta,\pi_2\rangle \in \mathsf{OneTps}^\varphi \times \mathsf{TwoTps}^+ \times \mathsf{OneTps}^\varphi$, by the Matching condition in Lemma 16, $\mathbf{g}_{\pi_1}(\eta,\pi_2) = \mathbf{g}_{\pi_2}(\widetilde{\eta},\pi_1)$. Thus $a_{\pi_1,\eta,\pi_2} = a_{\pi_2,\widetilde{\eta},\pi_1}$.
- (Non-triviality) By the Non-triviality condition in Lemma 16, there exists a 1-type $\pi \in \mathsf{OneTps}^\varphi$ such that $\mathbf{g}_\pi \neq \mathbf{0}$. Therefore,

$$\begin{aligned}\sum_{\pi_1\in\mathsf{OneTps}^\varphi}\sum_{\eta\in\mathsf{TwoTps}^+}\sum_{\pi_2\in\mathsf{OneTps}^\varphi} a_{\pi_1,\eta,\pi_2} &= \sum_{\pi_1\in\mathsf{OneTps}^\varphi}\sum_{\eta\in\mathsf{TwoTps}^+}\sum_{\pi_2\in\mathsf{OneTps}^\varphi} \mathbf{g}_{\pi_1}(\eta,\pi_2)\\ &\geq \sum_{\eta\in\mathsf{TwoTps}^+}\sum_{\pi_2\in\mathsf{OneTps}^\varphi} \mathbf{g}_\pi(\eta,\pi_2) \;>\; 0.\end{aligned}$$

Suppose that $\mathcal{Q}^\varphi(\mathbf{x})$ has a solution, assigning x_{π_1,η,π_2} to a_{π_1,η,π_2}. For every $\pi \in \mathsf{OneTps}^\varphi$, since $\mathbf{a}_\pi$ is a valid assignment of $\tilde{\mathcal{C}}^\varphi_\pi(\mathbf{x})$, by Lemma 3, $\mathbf{a}_\pi \in \mathsf{KleeneStar}(\mathcal{B}^\varphi_\pi)$. We claim that $\mathbf{a}_\pi$ are vectors satisfying the conditions in Lemma 16, which implies that φ is finitely satisfiable. The proof of Non-triviality is routine. Here we focus on the Matching constraint. For every $\langle \pi_1,\eta,\pi_2\rangle \in \mathsf{OneTps}^\varphi \times \mathsf{TwoTps}^+ \times \mathsf{OneTps}^\varphi$, by the Matching condition of $\mathcal{Q}^\varphi$, $a_{\pi_1,\eta,\pi_2} = a_{\pi_2,\widetilde{\eta},\pi_1}$. Since $\mathbf{a}_{\pi_1}(\eta,\pi_2) = a_{\pi_1,\eta,\pi_2}$, we have $\mathbf{a}_{\pi_1}(\eta,\pi_2) = \mathbf{a}_{\pi_2}(\widetilde{\eta},\pi_1)$. □

The system $\mathcal{Q}^\varphi(\mathbf{x})$ is a Boolean combination of exponentially many integer linear constraints, and thus can be solved in NEXP. To get EXP, we apply an idea from [17], refining to get exponentially many homogeneous constraints.

Definition 19. *For every* GP^2 *sentence* φ *in normal form, for* $M \in \mathbb{N}$*, the integer linear system* $\mathcal{Q}'^\varphi_M(\mathbf{x})$ *is the system* $\mathcal{Q}^\varphi(\mathbf{x})$ *where the Structure constraint is replaced with:*

- *(Structure′) For every* $\pi \in \mathsf{OneTps}^{\varphi}$,

  $$\exists \mathbf{y}_1, \mathbf{y}_2 \, (\mathbf{x}_\pi \; = \; \mathsf{proj}_n \, (\mathbf{y}_1) + \mathbf{y}_2) \wedge (\tilde{\mathbf{A}} \mathbf{y}_1 \; = \; \mathbf{0}) \wedge (\mathbf{A} \mathbf{y}_2 \; = \; \mathbf{0}) \wedge ((\mathbf{1} \cdot \mathbf{y}_2) \leq M(\mathbf{1} \cdot \mathbf{y}_2)),$$

  where $\tilde{\mathbf{A}}$ *is the matrix obtained by applying Lemma 3 to the characteristic system* $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$.

Lemma 20. *For every* GP^2 *sentence* φ *in normal form, there exists* $M^{\varphi} \in \mathbb{N}$*, such that for every* $M \geq M^{\varphi}$*,* $\mathcal{Q}^{\varphi}(\mathbf{x})$ *has a solution in* $\mathbb{N}$ *if and only if* $\mathcal{Q}'^{\varphi}_{M}(\mathbf{x})$ *has a solution in* $\mathbb{N}$*.*

Moreover, M^{φ} *can be computed directly from* φ *in time exponential in the length of* φ*, assuming binary encoding of coefficients.*

Proof. By Lemma 1, if $\mathcal{Q}^{\varphi}$ has a solution in $\mathbb{N}$, then it has a "small solution": every value is bounded by $M^{\varphi}_0 = c_1 t \left(t\widetilde{\lambda}\right)^{c_2 t}$, where $\widetilde{\lambda}$ is the maximal coefficient in φ and t is the number of constraints in $\mathcal{Q}^{\varphi}$, which is exponential in the number of unary and binary predicates. Thus M^{φ}_0 can be computed directly from φ in time exponential in the length of φ, assuming binary encoding of coefficients. Let $M^{\varphi} = 2^n M^{\varphi}_0$.

Note that the only difference between $\mathcal{Q}^{\varphi}(\mathbf{x})$ and $\mathcal{Q}'^{\varphi}_{M}(\mathbf{x})$ pertains to constraints $(\mathbf{y}_1 = \mathbf{0}) \rightarrow (\mathbf{y}_2 = \mathbf{0})$ and $((\mathbf{1} \cdot \mathbf{y}_2) \leq M(\mathbf{1} \cdot \mathbf{y}_1))$. We show that the two systems are equi-satisfiable, provided M is sufficiently large.

Suppose that $\mathcal{Q}^{\varphi}(\mathbf{x})$ **has a solution in** $\mathbb{N}$. We claim that a small solution $\mathbf{a}_\pi$ of $\mathcal{Q}^{\varphi}(\mathbf{x})$ is a solution of $\mathcal{Q}'^{\varphi}(\mathbf{x})$. For every $\pi \in \mathsf{OneTps}^{\varphi}$, let $\mathbf{b}_1$ and $\mathbf{b}_2$ be corresponding assignments for $\mathbf{y}_1$ and $\mathbf{y}_2$ in $\tilde{\mathcal{C}}^{\varphi}_{\pi}(\mathbf{x}_\pi)$. If $\mathbf{a}_\pi = \mathbf{0}$, then $\mathbf{b}_1 = \mathbf{0}$ and $\mathbf{b}_2 = \mathbf{0}$, which implies that $(\mathbf{1} \cdot \mathbf{b}_2) \leq M^{\varphi}(\mathbf{1} \cdot \mathbf{b}_1)$ holds trivially. On the other hand, if $\mathbf{a}_\pi \neq \mathbf{0}$, then $\mathbf{b}_1 \neq \mathbf{0}$, which implies that $(\mathbf{1} \cdot \mathbf{b}_1) \geq 1$. Therefore $(\mathbf{1} \cdot \mathbf{b}_2) \leq 2^n M^{\varphi}_0 = M^{\varphi} \leq M \leq M(\mathbf{1} \cdot \mathbf{b}_1)$

Suppose that $\mathcal{Q}'^{\varphi}_{M}(\mathbf{x})$ **has a solution** $\mathbf{a}_\pi$ **in** $\mathbb{N}$. We claim that the $\mathbf{a}_\pi$ also form a solution to $\mathcal{Q}^{\varphi}(\mathbf{x})$. It is sufficient to show that, for vectors $\mathbf{b}_1$ and $\mathbf{b}_2$, if $(\mathbf{1} \cdot \mathbf{b}_2) \leq M(\mathbf{1} \cdot \mathbf{b}_1)$, then $(\mathbf{b}_1 = \mathbf{0}) \rightarrow (\mathbf{b}_2 = \mathbf{0})$. If $\mathbf{b}_1 = \mathbf{0}$, then $(\mathbf{1} \cdot \mathbf{b}_2) \leq M(\mathbf{1} \cdot \mathbf{b}_1) = 0$, which implies that $\mathbf{b}_2 = \mathbf{0}$. Otherwise if $\mathbf{b}_1 \neq \mathbf{0}$, then $(\mathbf{b}_1 = \mathbf{0}) \rightarrow (\mathbf{b}_2 = \mathbf{0})$ holds trivially. □

We are now ready to prove Theorem 9:

Proof. For every GP^2 sentence φ in normal form, by Lemma 16, Lemma 18, and Lemma 20, φ is finitely satisfiable if and only if $\mathcal{Q}'^{\varphi}_{M^{\varphi}}(\mathbf{x})$ has a solution in $\mathbb{N}$. Note that $\mathcal{Q}'^{\varphi}_{M^{\varphi}}(\mathbf{x})$ is an integer linear system with $2^{\mathcal{O}(m(n+\log K))}$ variables, $2^{\mathcal{O}(m(n+\log K))}$ constraints, and coefficients bounded by M^{φ}. Moreover, $\mathcal{Q}'^{\varphi}_{M^{\varphi}}(\mathbf{x})$ is *homogeneous*. Thus, it has a solution in the natural numbers if and only if it has a *rational* solution. The feasibility of an integer linear system over the rationals can be checked in time polynomial in the number of variables, the number of equations, and the logarithm of the absolute value of the maximum coefficients [9]. Consequently, this process takes time exponential in the length of φ, assuming binary encoding of coefficients □

Note that the solutions to the linear system used in proving Theorem 9 will contain a representation of every finite model: the vector of cardinalities of 1-types. But it contains additional vectors that do not correspond to the cardinalities of one types in any model. This is related to the fact that while existence

of a pseudo-model implies existence of a model (Lemma 15), cardinalities are not preserved in moving from a pseudo-model to the corresponding model. We return to this in Section 6.

5 Spectra of C^2 formulas

In this section we will consider converting formulas in logic with arithmetic into linear constraints. These constraints will characterize the set of models of a C^2 formula. With this characterization in place, we have reduced finite satisfiability of the formula to a check for satisfiability of the conjunction of the linear constraints, which can be done via standard linear algebra solving. From this it will follow that C^2 with a variety of global cardinality constraints is decidable: we just conjoined the cardinality constraints with the constraints characterizing the models.

We fix a vocabulary σ which consists of n unary predicates $U_1, \ldots, U_n$ and m binary predicates $R_1, \ldots, R_m$. We consider C^2 sentence φ over σ in *normal form*:

$$\varphi \;:=\; \forall x\, \gamma(x) \wedge \forall x\, \forall y\; (x \neq y \rightarrow \alpha(x,y)) \wedge \bigwedge_{i \in [m']} \forall x\, \exists^{=k_i} y\; (R_i(x,y) \wedge x \neq y)\,,$$

where $\gamma(x)$ and $\alpha(x,y)$ are quantifier-free, R_i are binary predicates, $0 \leq m' \leq m$, and $k_i \in \mathbb{N}$.

Definition 21 (Count bound vector)**.** *The count bound vector of φ, denoted by $\mathbf{k}^\varphi$, is the m' dimension vector such that the ith component of $\mathbf{k}^\varphi$ is k_i.*

We denote the sum of k_i by K^φ.

Definition 22 (Forward and backward characteristic vector of 2-types)**.** *For m' as above and a 2-type η, the forward characteristic vector of η, denoted by $\eta^{\triangleright}$, is an m' dimensional vector defined: for every $1 \leq i \leq m'$, if $R_i(x,y) \in \eta$, then the ith component of $\eta^{\triangleright}$ is 1. Otherwise, it is 0. The backward characteristic vector of η, denoted by $\eta^{\triangleleft}$, is defined similarly.*

Definition 23 (Forward and backward silent 2-type)**.** *For m' as above and a 2-type η we say that η is forward-silent if for $1 \leq i \leq m'$, $\neg R_i(x,y) \in \eta$. Backward-silent is defined similarly.*

We define the notions of OneTps^φ and $\mathsf{TwoTps}^\varphi_{\pi_1,\pi_2}$ similarly to those in GP^2. We also define $\mathsf{TwoTps}^{\varphi,\mathsf{silent}}_{\pi_1,\pi_2} \subseteq \mathsf{TwoTps}^\varphi_{\pi_1,\pi_2}$ as the set of 2-types that are both forward- and backward-silent, $\mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2}$ for 2-types that are not forward-silent but are backward-silent, and $\mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2}$ for 2-types that are neither forward- nor backward-silent.

In the rest of the section, we deal only with normal form sentences. We explain how the results apply to arbitrary C^2 sentences in the appendix.

Definition 24 (Spectrum and 1-type spectrum)**.** *For every C^2 sentence φ, the spectrum of φ, denoted by $\mathsf{SPEC}(\varphi)$, is the set of natural numbers that are sizes of finite models of φ.*

The 1-type spectrum of φ, denoted by $\Pi\text{-}\mathsf{SPEC}(\varphi)$, is the set of 1-type cardinality vectors (see Definition 8) corresponding to finite models of φ.

We aim to show:

Theorem 25. *For every* C^2 *sentence* φ, $\mathsf{SPEC}(\varphi)$ *and* $\Pi\text{-}\mathsf{SPEC}(\varphi)$ *are semilinear. Furthermore, each spectrum can be represented by an existential Presburger formula of size doubly exponential in* φ.

Definition 26 (Silent compatible 1-types)**.** *For a pair of 1-types* π_1 *and* π_2*, we say that* π_1 *and* π_2 *are* silent-compatible *w.r.t.* φ*, if* $\mathsf{TwoTps}^{\varphi,\mathsf{silent}}_{\pi_1,\pi_2}$ *is non-empty. Otherwise, we say, following [16], that* π_1 *and* π_2 *are a* noisy pair*.*

Lemma 27. *For every* C^2 *sentence* φ *in normal form and* σ*-structure* $\mathcal{G} \models \varphi$*, for every pair of 1-types* π_1 *and* π_2*, if* π_1 *and* π_2 *are a noisy pair (w.r.t.* φ*), then at least one of the 1-types* π_1 *or* π_2 *is realized by at most* $2K^\varphi + 1$ *vertices in* $\mathcal{G}$*.*

Proof. Recall that V_π denotes the vertices in V with 1-type π. Because $\mathcal{G} \models \varphi$, for vertex $v \in \mathcal{G}$, $\sum_{u \in V \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) = \mathbf{k}^\varphi$. We consider two cases.

Suppose that $\pi_1 = \pi_2$**.** Since $\mathsf{TwoTp}^{\triangleright}(v,u) = \mathsf{TwoTp}^{\triangleleft}(u,v)$, we have

$$\begin{aligned}\sum_{v \in V_{\pi_1}} \sum_{u \in V_{\pi_1} \setminus \{v\}} (\mathsf{TwoTp}^{\triangleright}(v,u) + \mathsf{TwoTp}^{\triangleleft}(v,u)) &= 2 \sum_{v \in V_{\pi_1}} \sum_{u \in V_{\pi_1} \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) \\ &\preccurlyeq 2 \sum_{v \in V_{\pi_1}} \sum_{u \in V \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) \;=\; 2|V_{\pi_1}|\mathbf{k}^\varphi.\end{aligned}$$

For every pair of vertices v and u in V_{π_1} with $v \neq u$, by definition of noisy pair, the sum of components of $\mathsf{TwoTp}^{\triangleright}(v,u) + \mathsf{TwoTp}^{\triangleleft}(v,u)$ is at least 1. Therefore, we have

$$\sum_{v \in V_{\pi_1}} \sum_{u \in V_{\pi_1} \setminus \{v\}} 1 \;=\; |V_{\pi_1}| \cdot (|V_{\pi_1}| - 1) \;\leq\; 2|V_{\pi_1}| \cdot K^\varphi,$$

which implies that $|V_{\pi_1}| \leq 2K^\varphi + 1$. **Suppose that** $\pi_1 \neq \pi_2$**.** By a similar argument, $|V_{\pi_1}| \cdot |V_{\pi_2}| \leq (|V_{\pi_1}| + |V_{\pi_2}|) \cdot K^\varphi$. If $|V_{\pi_1}| \leq 2K^\varphi + 1$ we are done. So assume the opposite. Then we have

$$|V_{\pi_2}| \;\leq\; \frac{K^\varphi}{1 - \frac{K^\varphi}{|V_{\pi_1}|}} \;\leq\; \frac{K^\varphi}{1 - \frac{K^\varphi}{2K^\varphi+1}} \;\leq\; 2K^\varphi.$$

□

The following straightforward proposition gives conditions that characterize the satisfiability of C^2 by a model in terms of the prior constructs.

Proposition 28. *For every* C^2 *sentence* φ *in normal form and* σ*-structure* $\mathcal{G}$*,* $\mathcal{G}$ *is a model of* φ *if and only if following conditions hold. For every* $v, u \in V$ *with* $v \neq u$*,*

- *(1-type)* $\mathsf{OneTp}(v) \in \mathsf{OneTps}^\varphi$*,*
- *(2-type)* $\mathsf{TwoTp}(v,u) \in \mathsf{TwoTps}^\varphi_{\mathsf{OneTp}(v),\mathsf{OneTp}(u)}$*, and*
- *(Counting)* $\sum_{u \in V \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) = \mathbf{k}^\varphi$*,*

where $\mathbf{k}^\varphi$ *is the count bound vector of* φ*.*

Definition 29 (Partial model of C^2 sentence)**.** *For every* C^2 *sentence* φ *and* σ*-structure* $\mathcal{G}$*, we say that a complete colored graph* $\mathcal{G}$ *is a* partial model *of* φ *if* $\mathcal{G}$ *satisfies the first two conditions in Proposition 28.*

The following is obvious.

Lemma 30. *For every C^2 sentence φ in normal form, for every σ-structure $\mathcal{G} \models \varphi$, for every $\mathcal{H} \subseteq \mathcal{G}$, $\mathcal{H}$ is a partial model of φ.*

Core of structures.

Definition 31. *For $\mathcal{G}$ a σ-structure, $\ell \in \mathbb{N}$, we say that substructure $\mathcal{H} \subseteq \mathcal{G}$ is an ℓ-core of $\mathcal{G}$ if*

- *for every 1-type π, the number of vertices in $\mathcal{G} \setminus \mathcal{H}$ with 1-type π is either 0 or at least ℓ*
- *for 1-types π_1, π_2, and 2-type η, there are either 0 and at least ℓ pairs of vertices $v, u \in \mathcal{G} \setminus \mathcal{H}$ satisfying that $v \neq u$, $\mathsf{OneTp}(v) = \pi_1$, $\mathsf{TwoTp}(v, u) = \eta$, and $\mathsf{OneTp}(u) = \pi_2$.*

Note that the core of $\mathcal{G}$ is not unique, and $\mathcal{G}$ is trivially a core of itself for every $\ell \in \mathbb{N}$. Therefore, we are interested in the existence of a *small* core of $\mathcal{G}$. We can show that for every ℓ, every σ-structure has an *exponential-size* core.

Lemma 32. *For every σ-structure $\mathcal{G}$, for every $\ell \in \mathbb{N}$, $\mathcal{G}$ has an ℓ-core $\mathcal{H}$ with size at most $(2^{2n+4m+2}) \cdot \ell$.*

The intuition behind the construction is to repeatedly select a 1-type or tuple that violates the ℓ-core condition and add the relevant vertices to $\mathcal{H}$. Since there are only finitely many such choices, the process eventually stabilizes, and the size of the constructed core remains bounded: see the appendix for details.

For a C^2 sentence φ and a σ-structure $\mathcal{G} \models \varphi$, if $\mathcal{H}$ is a ℓ-core of $\mathcal{G}$ with sufficiently large ℓ, then we have the following important property of $\mathcal{G} \setminus \mathcal{H}$:

Lemma 33. *For every C^2 sentence φ and σ-structure $\mathcal{G} \models \varphi$, for every ℓ-core $\mathcal{H} \subseteq \mathcal{G}$ with $\ell > 2K^{\varphi} + 1$, the 1-types realized in $\mathcal{G} \setminus \mathcal{H}$ are silent-compatible (w.r.t. φ).*

Proof. Suppose that there exist 1-types π_1 and π_2 realized in $\mathcal{G} \setminus \mathcal{H}$ which are not silent-compatible. By Lemma 27, at least one of the 1-types π_1 or π_2 is realized by at most $2K^{\varphi} + 1$ vertices in $\mathcal{G}$. Without loss of generality, suppose that π_1 is realized by at most $2K^{\varphi} + 1$ vertices in $\mathcal{G}$, then there are at most $2K^{\varphi} + 1 < \ell$ vertices in $\mathcal{G} \setminus \mathcal{H}$ realized π_1. However, by the definition of ℓ-core, since π_1 is realized in $\mathcal{G} \setminus \mathcal{H}$, it is realized by at least ℓ vertices in $\mathcal{G} \setminus \mathcal{H}$. This leads to a contradiction. □

In Section 4, we introduced the notion of behavior vectors to describe the number of edges connected to a given vertex with specific 2-types. In this section, we consider σ-structures with a fixed core and extend the concept of behavior vectors to incorporate information about the core. Again, we fix an ordering of the set $\mathsf{TwoTps}^{+} \times \mathsf{OneTps}$, which has cardinality $2^{n+3m} - 2^{n+m}$.

Definition 34 (Extended behavior)**.** *Fixing a vocabulary σ, an* extended behavior *w.r.t. a set of vertices V is a tuple $\langle g, \mathbf{f} \rangle$ where g is a mapping from a vertex $v \in V$ to a 2-type in TwoTps, and $\mathbf{f}$ is an element of $\mathbb{N}^{2^{n+3m}-2^{n+m}}$.*

Informally, each extended behavior consists of two components. The first one is a mapping from vertices in the core to 2-types, capturing the 2-type realized

by the edge between a given vertex and core vertices. The second component is a behavior vector.

Abusing notation, for an extended behavior $\langle g, \mathbf{f}\rangle$, we will write $g^{\triangleright}(v)$ for $(g(v))^{\triangleright}$ and $\widetilde{g}(v)$ for the dual of $g(v)$. We will also write $\mathbf{f}(\eta, \pi)$ for the ith component of $\mathbf{f}$, where i is the index of the tuple $\langle \eta, \pi\rangle$ in the fixed ordering of $\mathsf{TwoTps}^{+} \times \mathsf{OneTps}$.

Definition 35 (Compatibility of extended behaviors)**.** *For a C^2 sentence φ in normal form, $\pi \in \mathsf{OneTps}^{\varphi}$, and a σ-structure $\mathcal{H}$ with vertices V, we say that an extended behavior function $\langle g, \mathbf{f}\rangle$ w.r.t. V is* compatible *with φ, π, and $\mathcal{H}$ if f satisfies:*

1. *(Compatibility with $\mathcal{H}$) For every $v \in V$, $g(v) \in \mathsf{TwoTps}^{\varphi}_{\pi,\mathsf{OneTp}(v)}$.*
2. *(Compatibility with 1- and 2- types) For every $\langle \eta, \pi'\rangle \in \mathsf{TwoTps}^{+} \times \mathsf{OneTps}$, if $\pi' \notin \mathsf{OneTps}^{\varphi}$ or $\eta \notin \mathsf{TwoTps}^{\varphi}_{\pi,\pi'}$, then $\mathbf{f}(\eta, \pi') = 0$.*
3. *(Forward-silent zero) For every $\langle \eta, \pi'\rangle \in \mathsf{TwoTps}^{+} \times \mathsf{OneTps}$, if η is forward-silent, then $\mathbf{f}(\eta, \pi') = 0$.*
4. *(Counting) The remaining entries of $\mathbf{f}$, that is, entries for 2-types that are not forward-silent, are bounded by the following constraint:*

$$\sum_{v \in V} g^{\triangleright}(v) + \sum_{\pi' \in \mathsf{OneTps}^{\varphi}} \left(\sum_{\eta \in \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi,\pi'}} \mathbf{f}(\eta, \pi') \cdot \eta^{\triangleright} + \sum_{\eta \in \mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi,\pi'}} \mathbf{f}(\eta, \pi') \cdot \eta^{\triangleright} \right) = \mathbf{k}^{\varphi}.$$

   Note that this condition implies that if π' and η are compatible with φ, and η is not forward-silent, then $\mathbf{f}(\eta, \pi')$ is bounded by K^{φ}.

Note that given an extended behavior, we can check whether it is compatible with π and $\mathcal{H}$ in time polynomial in 2^n and $|V|$. Let $\mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$ be the set of extended behaviors w.r.t. the vertices of $\mathcal{H}$ that are compatible with φ, π and $\mathcal{H}$. For each $\langle g, \mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$, recall that every entry of $\mathbf{f}$ is bounded by $|K^{\varphi}|$. Thus the cardinality of $\mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$ is bounded by $|V|^{|\mathsf{TwoTps}|} \cdot |K^{\varphi}|^{|\mathsf{TwoTps} \times \mathsf{OneTps}|}$ which is doubly exponential in the length of φ.

Computing a semilinear representation of the spectrum of a C^2 sentence. We are now ready to show that the 1-type spectrum of a C^2 sentence φ in normal form is semilinear. For every finite model $\mathcal{G}$ of φ, by Lemma 27, for sufficiently large ℓ, for every ℓ-core $\mathcal{H}$ of $\mathcal{G}$ and pair of 1-types π_1 and π_2 in $\mathcal{G} \setminus \mathcal{H}$, π_1 and π_2 are silent-compatible w.r.t. φ. Furthermore by Lemma 32, $\mathcal{G}$ has an ℓ-core whose size is exponential in φ and ℓ. We prove Theorem 37 via a two-step construction. First we show that for every such core $\mathcal{H}$, there exists an existential Presburger formula $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ such that the solution set of $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ corresponds to the set of 1-type cardinality vectors of finite models of φ with ℓ-core $\mathcal{H}$.

- The variables are x_π for $\pi \in \mathsf{OneTps}^{\varphi}$, and $y_{\pi,\langle g,\mathbf{f}\rangle}$ for $\pi \in \mathsf{OneTps}^{\varphi}$ and $\langle g, \mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$. Intuitively, $\mathbf{x}$ is the 1-type spectrum of the finite model $\mathcal{G}$ and $y_{\pi,\langle g,\mathbf{f}\rangle}$ is the number of vertices in $\mathcal{G} \setminus \mathcal{H}$ with 1-type π and extended behavior $\langle g, \mathbf{f}\rangle$.

- The formula $\mathsf{SUM}_{\mathcal{H}}(\mathbf{x},\mathbf{y})$ asserts that the 1-type spectrum of $\mathcal{G}$ is the sum of the 1-type spectrum of $\mathcal{H}$ and the 1-type spectrum of $\mathcal{G}\setminus\mathcal{H}$. Let s_π be the number of vertices in $\mathcal{H}$ with 1-type π.

  $$\mathsf{SUM}_{\mathcal{H}}(\mathbf{x},\mathbf{y}) := \bigwedge_{\pi\in\mathsf{OneTps}^{\varphi}} \left(x_\pi = s_\pi + \sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} y_{\pi,\langle g,\mathbf{f}\rangle} \right).$$

- The formula $\mathsf{COMP}_{\mathcal{H}}(\mathbf{y})$ guarantees that the vertices in $\mathcal{H}$ satisfies the Counting condition of Proposition 28. Let V_c be the vertices in $\mathcal{H}$.

  $$\mathsf{COMP}_{\mathcal{H}}(\mathbf{y}) := \bigwedge_{v\in V_c} \left(\sum_{u\in V_c\setminus\{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{\pi\in\mathsf{OneTps}^{\varphi}} \sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} g^{\triangleleft}(v)\cdot y_{\pi,\langle g,\mathbf{f}\rangle} = \mathbf{k}^{\varphi} \right).$$

- The formula $\mathsf{SILENT}(\mathbf{y})$ assert that 1-types realized in $\mathcal{G}\setminus\mathcal{H}$ are silent compatible.

  $$\mathsf{SILENT}_{\mathcal{H}}(\mathbf{y}) := \bigwedge_{\substack{\pi_1,\pi_2\in\mathsf{OneTps}^{\varphi}\\ \text{are not silent compatible}}} \left(\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} y_{\pi_1,\langle g,\mathbf{f}\rangle} = 0 \right) \vee \left(\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_2,\mathcal{H}}} y_{\pi_2,\langle g,\mathbf{f}\rangle} = 0 \right).$$

- The formula $\mathsf{BIG}_{\mathcal{H}}(\mathbf{y})$ guarantees that 1-types and tuples realized in $\mathcal{G}\setminus\mathcal{H}$ satisfy the requirements of a core.

  $$\mathsf{BIG}_{\mathcal{H}}(\mathbf{y}) := \bigwedge_{\pi\in\mathsf{OneTps}^{\varphi}} \exists z \left(z = \sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} y_{\pi,\langle g,\mathbf{f}\rangle} \right) \wedge \left(z = 0 \vee z \geq (2K^{\varphi}+1)^2 \right) \wedge$$

  $$\bigwedge_{\substack{\pi_1,\pi_2\in\mathsf{OneTps}^{\varphi}\\ \eta\in\mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2}}} \exists z \left(z = \sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2)\cdot y_{\pi_1,\langle g,\mathbf{f}\rangle} \right) \wedge \left(z = 0 \vee z \geq (2K^{\varphi}+1)^2 \right).$$

- The formula $\mathsf{MATCH}_{1,\mathcal{H}}(\mathbf{y})$ encodes the edge matching condition for audible 2-types and vertices in $\mathcal{G}\setminus\mathcal{H}$.

  $$\mathsf{MATCH}_{1,\mathcal{H}}(\mathbf{y}) := \bigwedge_{\substack{\pi_1,\pi_2\in\mathsf{OneTps}^{\varphi}\\ \eta\in\mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2}}} \left(\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2)\cdot y_{\pi_1,\langle g,\mathbf{f}\rangle} = \sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_2,\mathcal{H}}} \mathbf{f}(\widetilde{\eta},\pi_1)\cdot y_{\pi_2,\langle g,\mathbf{f}\rangle} \right).$$

- The formula $\mathsf{MATCH}_{2,\mathcal{H}}(\mathbf{y})$ encodes the edge matching condition for backward-silent but not forward-silent 2-types and vertices in $\mathcal{G}\setminus\mathcal{H}$.

  $$\mathsf{MATCH}_{2,\mathcal{H}}(\mathbf{y}) := \bigwedge_{\substack{\pi_1,\pi_2\in\mathsf{OneTps}^{\varphi}\\ \eta\in\mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2}}} \left(\left(\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2)\cdot y_{\pi_1,\langle g,\mathbf{f}\rangle} > 0 \right) \rightarrow \left(\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^{\varphi}_{\pi_2,\mathcal{H}}} y_{\pi_2,\langle g,\mathbf{f}\rangle} > 0 \right) \right).$$

  Keep in mind that we will always be considering solutions in the natural numbers. So this implication could be rewritten without summation: if for one of the extended behaviors for this triple, $\mathbf{f}(\eta,\pi_2)\cdot y_{\pi_1,\langle g,\mathbf{f}\rangle} > 0$, then one of the numbers $y_{\pi_2,\langle g,\mathbf{f}\rangle} > 0$.

Finally, $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x}) := \exists\mathbf{y}\ \mathsf{SUM}_{\mathcal{H}}(\mathbf{x},\mathbf{y}) \wedge \mathsf{COMP}_{\mathcal{H}}(\mathbf{y}) \wedge \mathsf{SILENT}_{\mathcal{H}}(\mathbf{y}) \wedge \mathsf{BIG}_{\mathcal{H}}(\mathbf{y}) \wedge \bigwedge_{i=1,2} \mathsf{MATCH}_{i,\mathcal{H}}(\mathbf{y})$.

This system characterizes the spectrum:

Lemma 36. *For every C^2 sentence φ and partial model $\mathcal{H}$ of φ, a vector $\mathbf{v}$ is a solution of $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ if and only if there exists a finite model $\mathcal{G}$ of φ such that the 1-type cardinality vector of $\mathcal{G}$ is $\mathbf{v}$, and $\mathcal{H}$ is a $(2K^{\varphi}+1)^2$-core of $\mathcal{G}$.*

It is easy to see that vectors in the spectrum satisfy the equations. In the other direction, we extend the core $\mathcal{H}$ to a finite model $\mathcal{G}$ of φ. Solutions to

the equations tell us how many vertices of each 1-type and extended behavior to create in $\mathcal{G}$. We assign edge colors so that each vertex realizes its intended extended behavior. This is always possible because the solution is sufficiently large, as ensured by the Big condition in the equation. This extra size gives us some leeway to match up 1-types via 2-types. Details are in the appendix.

Let $\mathfrak{H}$ be the collection of partial models of φ with size at most $2^{2n+4m+2} \cdot (2K^{\varphi}+1)^2$. We define the existential Presburger formula $\Psi^{\varphi}(\mathbf{x}) := \bigvee_{\mathcal{H}\in\mathfrak{H}} \Psi^{\varphi}_{\mathcal{H}}(\mathbf{y})$.

Lemma 37. *For every* C^2 *sentence* φ, $[\![\Psi^{\varphi}(\mathbf{x})]\!] = \Pi\text{-}\mathsf{SPEC}(\varphi)$.

Proof. Suppose that $\mathbf{v} \in [\![\Psi^{\varphi}(\mathbf{x})]\!]$, there exists $\mathcal{H}$ such that $\mathbf{v} \in [\![\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})]\!]$. By Lemma 36, there exists a model $\mathcal{G}$ of φ such that the 1-type cardinality vector of $\mathcal{G}$ is $\mathbf{v}$. Thus $\mathbf{v} \in \Pi\text{-}\mathsf{SPEC}(\varphi)$. On the other hand, suppose that $\mathbf{v} \in \Pi\text{-}\mathsf{SPEC}(\varphi)$. Then there is a model $\mathcal{G}$ of φ with 1-type cardinality vector $\mathbf{v}$. By Lemma 32, $\mathcal{G}$ has a $(2K^{\varphi}+1)^2$-core $\mathcal{H}$ with size at most $2^{2n+4m+2}(2K^{\varphi}+1)^2$, which implies that $\mathcal{H} \in \mathfrak{H}$. By Lemma 36, $\mathbf{v} \in [\![\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})]\!]$. Since $\mathcal{H} \in \mathfrak{H}$, $\mathbf{v} \in [\![\Psi^{\varphi}(\mathbf{x})]\!]$. □

Theorem 25 follows immediately from Lemma 37.

Note that the length of the formula $\Psi^{\varphi}(\mathbf{x})$ is doubly exponential in the length of φ, since the length of $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ is already doubly exponential in the length of φ, as the number of variables $y_{\pi,\langle g,\mathbf{f}\rangle}$ is doubly exponential. Moreover, there are doubly exponentially many possible cores $\mathcal{H}$.

Consequences for finite satisfiability of C^2 with global constraints. Note that in Lemma 37, we show that for every C^2 sentence φ in normal form, there exists an existential Presburger formula $\Psi^{\varphi}(\mathbf{x})$ of doubly exponential size such that $\Pi\text{-}\mathsf{SPEC}\varphi = [\![\Psi^{\varphi}(\mathbf{x})]\!]$. Since for every 1-type π and partial model $\mathcal{H}$ of φ, we can compute $\mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$ in doubly exponential time in the length of φ and $\mathcal{H}$ by a straightforward enumerate-and-check procedure, $\Psi^{\varphi}(\mathbf{x})$ can be computed from φ in doubly exponential time. Note that global constraints of C^2 sentences are linear constraints over the 1-type cardinality vector. Therefore the finite satisfiability problem of C^2 with global constraints reduces to satisfiability problem of $\Psi^{\varphi}(\mathbf{x})$ together with those linear constraints. This naive approach yields a 2-NEXP algorithm.

We observe that $\Psi^{\varphi}(\mathbf{x})$ can be rewritten as disjunctions of doubly exponentially many integer linear systems. Each system contains doubly exponentially many variables, but only *exponentially many constraints*. By Lemma 1, if such a system has a solution, then there exists a solution in which only exponentially many variables are assigned non-zero values.

Therefore, we can decide the finite satisfiability of C^2 with global constraints using the NEXP procedure in Algorithm 1.

We emphasize again that the argument here is completely generic, and the same comment holds to show that for *any* decidability extension of Presburger arithmetic (e.g. Büchi Arithemetic or Semenov Arithmetic [21]), C^2 with global cardinality constraints of that form is decidable, with the complexity the maximum of NEXP and the complexity of solving the corresponding constraints.

Algorithm 1 Algorithm for the finite satisfiability problem of C^2 with global constraints

1: **procedure** SAT(φ)
2: Compute $\mathsf{OneTps}^{\varphi}$ and $\mathsf{TwoTps}^{\varphi}_{\pi_1,\pi_2}$
3: Guess a exponential size partial model $\mathcal{H}$ of φ
4: Guess exponentially many non-zero variables $y'_{\pi,\langle g,\mathbf{f}\rangle}$
5: Construct the system $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ with only variables $y'_{\pi,\langle g,\mathbf{f}\rangle}$
6: Check the feasibility of $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ with global constraints
7: **end procedure**

6 Negative results about the spectrum of GP^2 sentences

The 1-type spectrum of a C^2 sentence is semilinear, and we showed how to calculate it in the previous section. For GP^2, earlier in the paper we were able to characterize models by a linear system: but the solutions to this linear system are not in one-to-one correspondence with the number of satisfiers of 1-types. Thus a natural question is whether the 1-type spectrum of a GP^2 sentence is semilinear. We answer this below in the negative.

Consider the following GP^2 formula φ over vocabulary $\{P_1, P_2, E\}$:

$$\varphi := \forall x\ ((\neg P_1(x) \wedge P_2(x)) \vee (P_1(x) \wedge \neg P_2(x))) \quad \wedge \bigwedge_{1\le i\le 2} \forall x\ P_i(x) \rightarrow \gamma_i(x)$$

$$\gamma_1(x) := \#_y\left[E(x,y) \wedge P_1(y)\right] - \#_y\left[E(x,y) \wedge P_2(y)\right] = 0$$

$$\gamma_2(x) := \#_y\left[E(y,x) \wedge P_1(y)\right] = 1.$$

Let $\mathcal{G}$ be a finite model of φ. A vertex in $\mathcal{G}$ is labeled with either P_1 or P_2. Let V_i be the set of vertices in $\mathcal{G}$ with label P_i. For every $v \in V_1$, the number of edges from v to V_1 is the same as the number of edges from v to V_2. For every vertex $v \in V_2$, there exists exactly one edge from V_1 to v. Therefore the cardinality of V_2 is the same as the number of edges between vertices of V_1, which is bounded by $|V_1| \cdot (|V_1| - 1)$. Thus the 1-type spectrum of φ is $\left\{(0, a, b, 0) \in \mathbb{N}^4 \middle|\ b \le a(a-1)\right\}$. Hence we have the following negative result.

Theorem 38. *The 1-type spectrum of a GP^2 sentence is not always semilinear.*

This does not resolve whether the set of cardinalities of finite models is a semilinear set, and it leaves open whether GP^2 with global cardinality constraints is decidable.

7 Related work

The logic GP^2 was shown decidable in [2, 12], while the decidability of C^2 with cardinality constraints is implicit in [11, 3]. Our complexity bound for GP^2 refines the analysis of the Kleene star operator in [4], which in turn relies on [8]. Our analysis of C^2 with constraints obviously relies heavily on the techniques developed by Pratt-Hartmann for analysis of C^2, in particular normal forms, the

categorization of types, and the distinction between large and small number of realizers (e.g. the notion of Z-differentiated [16, 17]).

Although we borrow techniques from prior work, our approach gives a self-contained proof of decidability for C^2 itself, and also for C^2 with global constraints in extensions of Presburger arithmetic. An alternative approach to get bounds for C^2 with Presburger global constraints is by reducing to decidability of C^2. This approach was presented by Rudolph in [19], relying on ideas from [1]. Although technically speaking the results in [19] are presented for description logics, they could easily be applied to give the same bounds for C^2.

In addition to contributing tight bounds for complexity for decidability of GP^2 a well as the size of the spectrum for C^2, we note that [4] is phrased in terms of graph neural networks, while from the analysis of spectra in [11, 3] it would be difficult to see how semilinear representations are derived. Thus we believe our work additionally makes the analyses of logics with arithmetic substantially more accessible.

8 Conclusion

In the paper we isolate the complexity of satisfiability for two logics combining relational structure and arithmetic, GP^2 and the logics C^2 and GC^2 extended with cardinality constraints. In the process we refine and simplify proof techniques for analyzing such logics. GP^2 and C^2 are incomparable in expressiveness. The former allows inductively forming open formulas that involve arithmetic between counts of neighbors, while in the latter arithmetic can only be used at top-level for forming sentences. On the other hand, C^2 allows unguarded formulas. See the appendix for a formal argument. We leave open the question of the decidability of combinations of formulas in the two languages.

9 Proof of Lemma 3: computation of a representation of the Kleene closure

Recall that one important component in the decidability of GP^2 is the computation of a representation of the Kleene closure of a semilinear set. Recall from Section 2 that $\mathsf{proj}_n(\mathbf{v})$ denotes the projection operator that maps a vector $\mathbf{v}$ to its first n entries. The formal statement was:

For every integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$ with $\mathbf{c} \neq \mathbf{0}$, where $\mathbf{A} \in \mathbb{Z}^{m\times n}$ and $\mathbf{c} \in \mathbb{Z}^m$, there exists $\tilde{\mathbf{A}} \in \mathbb{Z}^{(n+t)\times(n+k)}$ with $\left\|\tilde{\mathbf{A}}\right\| = 1$ such that

$$\mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right) \;=\; \left\llbracket \tilde{\mathcal{Q}}(\mathbf{x}) \right\rrbracket,$$

where $\tilde{\mathcal{Q}}(\mathbf{x})$ is the following Boolean combination of integer linear systems

$$\exists \mathbf{y}_1, \mathbf{y}_2 \left(\mathbf{x} \;=\; \mathsf{proj}_n(\mathbf{y}_1) + \mathbf{y}_2\right) \wedge \left(\tilde{\mathbf{A}}\mathbf{y}_1 \;=\; \mathbf{0}\right) \wedge \left(\mathbf{A}\mathbf{y}_2 \;=\; \mathbf{0}\right) \wedge \left((\mathbf{y}_1 = \mathbf{0}) \rightarrow (\mathbf{y}_2 = \mathbf{0})\right),$$

$k = n \cdot (2D+1)^m + 1$, $t \leq 2nk$, and $D = n \cdot \|\mathbf{A}\| \cdot ((n+1) \cdot \|\mathbf{A}\| + \|\mathbf{c}\| + 1)^m$. Moreover, $\tilde{\mathbf{A}}$ can be computed in time $2^{\mathcal{O}\left(\log n + \log K + m^2\right)}$, where $K := \max\left(\|\mathbf{A}\|, \|\mathbf{c}\|\right)$.

We prove the lemma by considering *minimal solutions* of the integer linear system. Recall that $\preccurlyeq$ means pointwise inequality between vectors.

Definition 39 (Minimal elements)**.** *For a set $\mathcal{S} \subseteq \mathbb{N}^n$ and $\mathbf{v} \in \mathcal{S}$, we say that $\mathbf{v}$ is* minimal *in $\mathcal{S}$ if, for every $\mathbf{u} \in \mathcal{S}$ with $\mathbf{u} \neq \mathbf{v}$, $\mathbf{u} \not\leqslant \mathbf{v}$.*

Note that the following properties about minimal elements are trivial.

Lemma 40. *Let $\mathcal{S}$ be a nonempty subset of $\mathbb{N}^n$.*

1. *There exists $\mathbf{v} \in \mathcal{S}$ such that $\mathbf{v}$ is minimal in $\mathcal{S}$.*
2. *For every $\mathbf{v} \in \mathcal{S}$, there exists a minimal element $\mathbf{u} \in \mathcal{S}$, such that $\mathbf{v} - \mathbf{u} \in \mathbb{N}^n$.*

For an integer linear system $\mathcal{Q}(\mathbf{x})$, let $[\![\mathcal{Q}(\mathbf{x})]\!]_{\min}$ denotes the set of minimal solutions, i.e., $[\![\mathcal{Q}(\mathbf{x})]\!]_{\min} := \{\mathbf{v} \in [\![\mathcal{Q}(\mathbf{x})]\!] |\ \mathbf{v} \text{ is minimal in } [\![\mathcal{Q}(\mathbf{x})]\!]\}$. Note that the maximal entry of minimal solutions is bounded.

Lemma 41 ([15], also see Proposition 4. in [5])**.** *For every integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$, where $\mathbf{A} \in \mathbb{Z}^{m \times n}$ and $\mathbf{c} \in \mathbb{Z}^m$, for every $\mathbf{v} \in [\![\mathcal{Q}(\mathbf{x})]\!]_{min}$, $\|\mathbf{v}\| \leq ((n+1) \cdot \|\mathbf{A}\| + \|\mathbf{c}\| + 1)^m$.*

We begin by turning to the automata-theoretic approach for the solutions of integer linear systems.

Definition 42 (Acyclic, unambiguous, and simple finite automata)**.** *As usual, a finite automaton can be viewed as a directed graph where the vertices are the states and the edges are the transitions.*

Such an automaton is acyclic *if the graph does not contain a cycle. It is* unambiguous *if every word has at most one accepting run. We say that a finite automaton is* simple *if it is unambiguous, acyclic, and has exactly one accepting state.*

Definition 43 (Parikh Vector)**.** *For a word $w = a_{t_1} \cdots a_{t_\ell} \in \Sigma^*$, the* Parikh vector *of w is a vector in $\mathbb{N}^n$, defined as $\mathsf{Parikh}(w) := \sum_{i \in [\ell]} \mathbf{e}_{t_i}$. Intuitively, the ith entry in $\mathsf{Parikh}(w)$ is the number of occurrences of a_i in w.*

For a finite automaton $\mathcal{M}$, the Parikh image *of $\mathcal{M}$ is defined as*

$$\mathsf{Parikh}(\mathcal{M}) \ := \ \{\mathsf{Parikh}(w) |\ \mathcal{M} \text{ accepts } w\}.$$

We can encode a finite subset of solutions of a linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$ as the Parikh image of a finite automaton $\mathcal{M}$. More precisely, $\mathsf{Parikh}(\mathcal{M})$ contains all the minimal solutions of $\mathcal{Q}(\mathbf{x})$. The main idea behind the construction of $\mathcal{M}$ is as follows. Suppose $\mathcal{Q}(\mathbf{x})$ has a solution and $\mathbf{A} \in \mathbb{Z}^{m \times n}$. By Lemma 41, the components of every minimal solution are bounded above by $L := ((n+1) \cdot \|\mathbf{A}\| + \|\mathbf{c}\| + 1)^m$. We can construct a finite automaton $\mathcal{M}$ where the set of states is $\{\mathbf{A}\mathbf{v} |\ \mathbf{v} \in [0, L]^n\}$ and the transitions are $\langle \mathbf{A}\mathbf{v}, a_i, \mathbf{A}(\mathbf{v} + \mathbf{e}_i) \rangle$ for every $1 \leq i \leq n$. The initial state is $\mathbf{0}$ and the accepting state is $\mathbf{c}$. Obviously, $\mathsf{Parikh}(\mathcal{M})$ contains all the minimal solutions. In fact, it contains all solutions that are bounded by L. Note also that $\mathcal{M}$ can be constructed in exponential time. With a little more work, we can make $\mathcal{M}$ simple, a property that will be useful when analyzing the Kleene star of $[\![\mathcal{Q}]\!]$.

Lemma 44. *For every integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$, where $\mathbf{A} \in \mathbb{Z}^{m\times n}$ and $\mathbf{c} \in \mathbb{Z}^m$, there exists a simple finite automaton $\mathcal{M}$ with k states and t transitions, where $k = n \cdot L \cdot (2D+1)^m + 1$, $t \leq 2nk$, $L = ((n+1) \cdot \|\mathbf{A}\| + \|\mathbf{c}\| + 1)^m$ and $D = n \cdot \|\mathbf{A}\| \cdot L$, such that*

$$[\![\mathcal{Q}(\mathbf{x})]\!]_{min} \subseteq \mathsf{Parikh}(\mathcal{M}) \subseteq [\![\mathcal{Q}(\mathbf{x})]\!].$$

Moreover, $\mathcal{M}$ can be constructed in time exponential in $n, m, \|\mathbf{A}\|, \|\mathbf{c}\|$.

Proof. If $\mathbf{c} = \mathbf{0}$, then the system $\mathcal{Q}(\mathbf{x})$ admits the solution $\mathbf{0}$. The finite automaton that accepts only the empty word is the desired $\mathcal{M}$. So we concentrate on the case when $\mathbf{c} \neq \mathbf{0}$.

The automaton $\mathcal{M}$ is very similar to the one described in the intuition preceding the lemma. We fix a finite alphabet of n symbols $\Sigma := \{a_1, \ldots, a_n\}$. To ensure acyclicity, we modify the automaton $\mathcal{M}$ to accept only words from $a_1^* a_2^* \cdots a_n^*$. The automaton will also implement a counter that counts the number of occurrences of each a_i up to L. Such a counter can be implemented using the states of $\mathcal{M}$. We now give the details.

Let $\mathcal{D}$ be the set $[-D,\ D]^m$.

- The set of states is $\{q_{acc}\} \cup ([n] \times [L] \times \mathcal{D})$.
- The initial state is $(0, 0, \mathbf{0})$.
- For each $0 \leq i < j \leq n$, $0 \leq t \leq L-1$, and $\mathbf{v} \in \mathcal{D}$, we have the transitions:
  - if $\mathbf{v} + \mathbf{A}\mathbf{e}_i \in \mathcal{D}$, then $(\langle i, t, \mathbf{v}\rangle, a_i, \langle i, t+1, \mathbf{v} + \mathbf{A}\mathbf{e}_i\rangle) \in \Delta$,
  - if $\mathbf{v} + \mathbf{A}\mathbf{e}_i = \mathbf{c}$, then $(\langle i, t, \mathbf{v}\rangle, a_i, q_{acc}) \in \Delta$.
  - if $\mathbf{v} + \mathbf{A}\mathbf{e}_j \in \mathcal{D}$, then $(\langle i, t, \mathbf{v}\rangle, a_j, \langle j, 1, \mathbf{v} + \mathbf{A}\mathbf{e}_j\rangle) \in \Delta$,
  - if $\mathbf{v} + \mathbf{A}\mathbf{e}_j = \mathbf{c}$, then $(\langle i, t, \mathbf{v}\rangle, a_j, q_{acc}) \in \Delta$.

Obviously, $\mathcal{M}$ has only one accepting state. The automaton $\mathcal{M}$ is acyclic since it can only go from the state $\langle i, t, \mathbf{v}\rangle$ to $\langle j, t', \mathbf{u}\rangle$ when $i < j$ or when $i = j$ and $t < t'$. It is also unambiguous since for every state $\langle i, t, \mathbf{v}\rangle$ and $a \in \Sigma$, there is at most one state $\langle i', t', \mathbf{u}\rangle$ such that $\langle i, t, \mathbf{v}\rangle, a, \langle i', t', \mathbf{u}\rangle$ is a transition. Finally, it is routine to verify that $\mathsf{Parikh}(\mathcal{M}) \subseteq [\![\mathcal{Q}]\!]$. That $[\![\mathcal{Q}]\!]_{\min} \subseteq \mathsf{Parikh}(\mathcal{M})$ follows from Lemma 41. □

Next, we consider the Kleene star of sets. Recall the definition:

For every $\mathcal{S} \subseteq \mathbb{N}^n$, the *Kleene star* of $\mathcal{S}$ is defined as

$$\mathsf{KleeneStar}(\mathcal{S}) := \left\{ \sum_{s \in \mathcal{S}'} s \;\middle|\; \mathcal{S}' \text{ is a finite multisubset of } \mathcal{S} \right\}.$$

In the following we fix a finite alphabet of n symbols $\Sigma := \{a_1, \ldots, a_n\}$ and assume that every finite automaton is over this alphabet. We need more terminology concerning finite automata. Note that for every acyclic finite automaton $\mathcal{M}$ (Definition 42) a run r of $\mathcal{M}$ uses each transition at most once. Therefore, we can represent r by a Boolean vector with dimension $|\Delta|$. Recall that for a unambiguous finite automaton $\mathcal{M}$ and an accepted word w, there exists exactly one accepting run of $\mathcal{M}$ on w. Let $\mathbb{B} := \{0, 1\}$.

Definition 45. *For every simple finite automaton $\mathcal{M}$ with t transitions, for every accepted word w of $\mathcal{M}$, letting r be the unique accepting run of $\mathcal{M}$ on w, the* path vector *of w, denoted by $\mathsf{Path}(w)$, is the vector in $\mathbb{B}^t$ defined as follows: for $1 \leq i \leq t$, the ith entry of $\mathsf{Path}(w)$ is 1 if and only if r contains the ith transition.*

Lemma 46. *For every simple finite automaton $\mathcal{M}$ with t transitions, there exists a matrix $\mathbf{B}_{\mathcal{M}} \in \mathbb{Z}^{n \times t}$, which can be computed in time $\mathcal{O}(nt)$, such that*

- $\|\mathbf{B}_{\mathcal{M}}\| = 1$, *and*
- *for every word w accepted by $\mathcal{M}$, $\mathsf{Parikh}(w) = \mathbf{B}_{\mathcal{M}} \cdot \mathsf{Path}(w)$.*

Proof. We construct the matrix $\mathbf{B}$ as follows. For $1 \leq i \leq n$ and $1 \leq j \leq t$, if the tth transition is of the form (q, a_i, p), then $(B_{\mathcal{M}})_{i,j} = 1$. Otherwise, $(B_{\mathcal{M}})_{i,j} = 0$. It is trivial to verify that $\mathsf{Parikh}(w) = \mathbf{B} \cdot \mathsf{Path}(w)$. □

We now prove a lemma regarding the Kleene closure of the Parikh image. The proof employs a similar idea as in the argument for Theorem 1 in [20].

Lemma 47. *For every simple finite automaton $\mathcal{M}$ with k states $q_1, \ldots, q_k$ and t transitions, there exists a matrix $\mathbf{A}_{\mathcal{M}} \in \mathbb{Z}^{k \times t}$, which can be computed in time $\mathcal{O}(kt)$, such that*

- $\|\mathbf{A}_{\mathcal{M}}\| = 1$, *and*
- $\mathsf{KleeneStar}(\mathsf{Parikh}(\mathcal{M})) = [\![\mathcal{Q}_{\mathcal{M}}(\mathbf{x})]\!]$, *where* $\mathcal{Q}_{\mathcal{M}}(\mathbf{x}) := \exists \mathbf{y}\, (\mathbf{x} = \mathsf{proj}_n(\mathbf{y})) \wedge (\tilde{\mathbf{A}}_{\mathcal{M}} \mathbf{y} = \mathbf{0})$ *and* $\tilde{\mathbf{A}}_{\mathcal{M}} = \left[\begin{array}{c|c} -\mathbf{I} & \mathbf{B}_{\mathcal{M}} \\ \hline \mathbf{0} & \mathbf{A}_{\mathcal{M}} \end{array}\right]$.

Proof. We construct the matrix $\mathbf{A}_{\mathcal{M}}$ as follows. For $1 \leq i \leq k$ and $1 \leq j \leq t$, if q_i is the initial state or the accepting state, then $(A_{\mathcal{M}})_{i,j} = 0$. Otherwise,

$$(A_{\mathcal{M}})_{i,j} := \begin{cases} 1, & \text{if the } j^{th} \text{ transition is of the form } (p, a, q_i) \\ -1, & \text{if the } j^{th} \text{ transition is of the form } (q_i, a, p) \\ 0, & \text{otherwise.} \end{cases}$$

Note that since $\mathcal{M}$ is acyclic, there is no transition of the form (q_i, a, q_i).

We first claim that for every accepted word w, we have

$$\mathbf{A}_{\mathcal{M}} \cdot \mathsf{Path}(w) = \mathbf{0}.$$

We will show that $(\mathbf{A}_{\mathcal{M}} \cdot \mathsf{Path}(w))_i = 0$, for $1 \leq i \leq k$. If q_i is the initial state or the accepting state, by definition, $(A_{\mathcal{M}})_{i,j} = 0$ for every $1 \leq j \leq k$, hence, $(\mathbf{A}_{\mathcal{M}} \cdot \mathsf{Path}(w))_i = 0$.

If q_i does not appear in the accepting run of w, then $\mathsf{Path}(w)_j = 0$ if the jth transition is going into or coming out of state q_i. Hence, $(\mathbf{A}_{\mathcal{M}} \cdot \mathsf{Path}(w))_i = 0$.

Now, suppose that q_i appears in $\mathsf{Path}(w)$ and it is neither the initial state nor the accepting state. Let r be the accepting run of $\mathcal{M}$ on w. Since $\mathcal{M}$ is acyclic, q_i appears only once, which implies that r is of the form

$$\cdots p \xrightarrow{a} q_i \xrightarrow{a'} p' \cdots$$

Suppose that $\langle p, a, q_i \rangle$ is the j_1^{th} transition and $\langle q_i, a', p' \rangle$ is the j_2^{th} transition. Then we have:

$$(\mathbf{A}_{\mathcal{M}} \cdot \mathsf{Path}(w))_i = (\mathbf{A}_{\mathcal{M}})_{i,j_1} + (\mathbf{A}_{\mathcal{M}})_{i,j_2} = 1 + (-1) = 0.$$

We will now show that $\mathsf{KleeneStar}(\mathsf{Parikh}(\mathcal{M})) = [\![\mathcal{Q}_{\mathcal{M}}(\mathbf{x})]\!]$.

$\underline{\mathsf{KleeneStar}(\mathsf{Parikh}(\mathcal{M})) \subseteq [\![\mathcal{Q}_{\mathcal{M}}(\mathbf{x})]\!]}$**.** Suppose that $\mathbf{v} \in \mathsf{KleeneStar}(\mathsf{Parikh}(\mathcal{M}))$. By the definition of Kleene star and Parikh image, there exist accepted words $w_1, \ldots, w_\ell$ and $\mathbf{v} = \sum_{i \in [\ell]} \mathsf{Parikh}(w_i)$. Let $\mathbf{u} := \sum_{i \in [\ell]} \mathsf{Path}(w_i)$. By the claim above and Lemma 46, we have

$$\begin{aligned} \mathbf{A}_{\mathcal{M}}\mathbf{u} &= \sum_{i \in [\ell]} \mathbf{A} \cdot \mathsf{Path}(w_i) = \mathbf{0} \\ \mathbf{B}_{\mathcal{M}}\mathbf{u} &= \sum_{i \in [\ell]} \mathbf{B} \cdot \mathsf{Path}(w_i) = \sum_{i \in [\ell]} \mathsf{Parikh}(w_i) = \mathbf{v}, \end{aligned}$$

which implies that concatenation of $\mathbf{v}$ and $\mathbf{u}$ is a solution of the integer linear system $\mathcal{Q}_{\mathcal{M}}(\mathbf{x})$. Hence, $\mathbf{v} \in [\![\mathcal{Q}_{\mathcal{M}}(\mathbf{x})]\!]$.

$\underline{[\![\mathcal{Q}_{\mathcal{M}}(\mathbf{x})]\!] \subseteq \mathsf{KleeneStar}(\mathsf{Parikh}(\mathcal{M}))}$**.** Suppose that $\mathbf{v} \in [\![\mathcal{Q}_{\mathcal{M}}(\mathbf{x})]\!]$. Let $\mathbf{u}$ be the vector satisfying $\mathbf{v} = \mathbf{B}_{\mathcal{M}}\mathbf{u}$ and $\mathbf{A}_{\mathcal{M}}\mathbf{u} = \mathbf{0}$. We interpret the jth entry of $\mathbf{u}$, denoted by u_j as a flow of the jth edge (transition) in the transition diagram of $\mathcal{M}$. Then, since $\mathbf{A}_{\mathcal{M}}\mathbf{u} = \mathbf{0}$ by the claim above, for every node (state) q, if q is neither the initial state nor the accepting state, then

$$\sum_{\substack{j \in [t] \\ \text{the } j^{th} \text{ transition is of the form } (p,a,q)}} u_j - \sum_{\substack{j \in [t] \\ \text{the } j^{th} \text{ transition is of the form } (q,a,p)}} u_j = 0.$$

That is, the incoming flow to the node (state) q equals its outgoing flow. By the flow decomposition theorem [7], there exist paths (words) $w_1, \ldots, w_\ell$ from the initial state to the accepting state such that $\mathbf{u} = \sum_{i \in [\ell]} \mathsf{Path}(w_i)$. Since each w_i is a path from the initial state to the accepting state, each w_i is a word accepted by $\mathcal{M}$. By Lemma 46, for each $i \leq \ell$, $\mathsf{Parikh}(w_i) = \mathbf{B}_{\mathcal{M}} \cdot \mathsf{Path}(w_i)$. Thus, we have

$$\mathbf{v} = \mathbf{B}_{\mathcal{M}}\mathbf{u} = \sum_{i \in [\ell]} \mathbf{B}_{\mathcal{M}} \cdot \mathsf{Path}(w_i) = \sum_{i \in [\ell]} \mathsf{Parikh}(w_i) \in \mathsf{KleeneStar}(\mathsf{Parikh}(\mathcal{M})).$$

□

Combining Lemma 44 and Lemma 47 we obtain a lemma regarding the Kleene star of the set of minimal solutions of an integer linear system.

Lemma 48. *For every integer linear system $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$, where $\mathbf{A} \in \mathbb{Z}^{m \times n}$, $\mathbf{c} \in \mathbb{Z}^m$, there exists $\tilde{\mathbf{A}} \in \mathbb{Z}^{(n+t) \times (n+k)}$ with $\left\|\tilde{\mathbf{A}}\right\| = 1$ such that*

$$\mathsf{KleeneStar}\left([\![\mathcal{Q}(\mathbf{x})]\!]_{min}\right) \subseteq \left[\!\left[\exists \mathbf{y}\ (\mathbf{x} = \mathsf{proj}_n(\mathbf{y})) \wedge (\tilde{\mathbf{A}}\mathbf{y} = \mathbf{0})\right]\!\right] \subseteq \mathsf{KleeneStar}([\![\mathcal{Q}(\mathbf{x})]\!]),$$

where $k = n \cdot (2D+1)^m + 1$, $t \leq 2nk$, and $D = n \cdot \|\mathbf{A}\| \cdot ((n+1) \cdot \|\mathbf{A}\| + \|\mathbf{c}\| + 1)^m$. Moreover, $\tilde{\mathbf{A}}$ can be computed in time $2^{\mathcal{O}\left(\log n + \log K + m^2\right)}$, where $K := \max(\|\mathbf{A}\|, \|\mathbf{c}\|)$.

We are now ready to prove Lemma 3.

Proof. Let $\mathcal{Q}(\mathbf{x}) : \mathbf{A}\mathbf{x} = \mathbf{c}$. Let $\tilde{\mathbf{A}}$ be the matrix obtained by applying Lemma 48 to the integer linear system $\mathcal{Q}(\mathbf{x})$. It is routine to verify that the dimension and maximal absolute values of the entries of $\tilde{\mathbf{A}}$ satisfy the required bounds, and $\tilde{\mathbf{A}}$ can be computed within the desired time complexity.

Here we show that $\mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right) = \left\llbracket \tilde{\mathcal{Q}}(\mathbf{x}) \right\rrbracket$, where

$$\tilde{\mathcal{Q}}(\mathbf{x}) \ := \ \exists \mathbf{y}_1, \mathbf{y}_2 \left(\mathbf{x} \ = \ \mathsf{proj}_n\left(\mathbf{y}_1\right) + \mathbf{y}_2\right) \wedge \left(\tilde{\mathbf{A}}\mathbf{y}_1 \ = \ \mathbf{0}\right) \wedge \left(\mathbf{A}\mathbf{y}_2 \ = \ \mathbf{0}\right) \wedge \left(\left(\mathbf{y}_1 = \mathbf{0}\right) \rightarrow \left(\mathbf{y}_2 = \mathbf{0}\right)\right).$$

$\underline{\mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right) \subseteq \left\llbracket \tilde{\mathcal{Q}}(\mathbf{x}) \right\rrbracket}$. Suppose that $\mathbf{v} \in \mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right)$. If $\mathbf{v} = \mathbf{0}$, then we can choose $\mathbf{0}$ as the assignment for $\mathbf{y}_1$ and $\mathbf{y}_1$. Otherwise if $\mathbf{v} \neq \mathbf{0}$, by definition of Kleene star and minimal elements, there exists $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k \in \llbracket \mathcal{Q}(\mathbf{x}) \rrbracket$ and $\mathbf{u}_{1,\min}, \mathbf{u}_{2,\min}, \ldots, \mathbf{u}_{k,\min} \in \llbracket \mathcal{Q}(\mathbf{x}) \rrbracket_{\min}$ such that $\mathbf{v} = \sum_{i \in [k]} \mathbf{u}_i$, and for $1 \leq i \leq k$, $\left(\mathbf{u}_i - \mathbf{u}_{i,\min}\right) \in \mathbb{N}^n$.

- Because $\sum_{i \in [k]} \mathbf{u}_{i,\min} \in \mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket_{\min}\right)$, by Lemma 48, there exists $\mathbf{v}_1$ such that $\sum_{i \in [k]} \mathbf{u}_{i,\min} = \mathsf{proj}_n\left(\mathbf{v}_1\right)$ and $\tilde{\mathbf{A}}\mathbf{v}_1 = \mathbf{0}$.
- Let $\mathbf{v}_2 = \sum_{i \in [k]} (\mathbf{u}_i - \mathbf{u}_{i,\min})$. Observe that $\mathbf{A}\mathbf{v}_2 = \sum_{i \in [k]} \left(\mathbf{A}\mathbf{u}_i - \mathbf{A}\mathbf{u}_{i,\min}\right) = \mathbf{0}$.
- Because $\mathbf{c} \neq \mathbf{0}$, $\mathbf{0}$ is not a solution of $\mathcal{Q}(\mathbf{x})$, we have $\mathbf{v}_1 \neq \mathbf{0}$. Hence $\left(\mathbf{v}_1 = \mathbf{0}\right) \rightarrow \left(\mathbf{v}_2 = \mathbf{0}\right)$ holds.

Since $\mathbf{v} = \mathsf{proj}_n\left(\mathbf{v}_1\right) + \mathbf{v}_2$, $\mathbf{v}$ is a solution of $\tilde{\mathcal{Q}}$, where the solution assigns $\mathbf{y}_1$ to $\mathbf{v}_1$ and assigns $\mathbf{y}_2$ to $\mathbf{v}_2$.

$\underline{\left\llbracket \tilde{\mathcal{Q}}(\mathbf{x}) \right\rrbracket \subseteq \mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right)}$. Suppose that $\mathbf{v}$ is a solution of $\tilde{\mathcal{Q}}(\mathbf{x})$ by assigning $\mathbf{v}_1$ to $\mathbf{y}_1$ and assigning $\mathbf{v}_2$ to $\mathbf{y}_2$.

- If $\mathbf{v}_1 = \mathbf{0}$, then $\mathbf{v}_2 = \mathbf{0}$, which implies that $\mathbf{v} = \mathsf{proj}_n\left(\mathbf{v}_1\right) + \mathbf{v}_2 = \mathbf{0}$. By definition of Kleene star, $\mathbf{0} \in \mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right)$.
- Otherwise if $\mathbf{v}_1 \neq \mathbf{0}$, by Lemma 48, there exists $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k \in \llbracket \mathcal{Q}(\mathbf{x}) \rrbracket$ such that $\mathsf{proj}_n\left(\mathbf{v}_1\right) = \sum_{i \in [k]} \mathbf{u}_i$. Note that because $\mathbf{A}\mathbf{v}_2 = \mathbf{0}$, we have

  $$\mathbf{A}(\mathbf{u}_1 + \mathbf{v}_2) \ = \ \mathbf{A}\mathbf{u}_1 + \mathbf{A}\mathbf{v}_2 \ = \ \mathbf{c} + \mathbf{0} \ = \ \mathbf{c},$$

  which implies that $\mathbf{u}_1 + \mathbf{v}_2$ is a solution of $\mathcal{Q}(\mathbf{x})$. Therefore, we infer

  $$\mathbf{v}_1 + \mathbf{v}_2 \ = \ (\mathbf{u}_1 + \mathbf{v}_2) + \sum_{i \in [2,\ k]} \mathbf{u}_i \ \in \ \mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right).$$

  The membership in $\mathsf{KleeneStar}\left(\llbracket \mathcal{Q}(\mathbf{x}) \rrbracket\right)$ follows from the fact that $\mathbf{u}_1 + \mathbf{v}_2, \mathbf{u}_2, \ldots, \mathbf{u}_k$ are all solutions of $\mathcal{Q}$.

□

10 From normal form sentences to general sentences

In the main body all our results are proven for sentences in normal form. In this appendix we will show that they also hold for general sentences. In particular

we will sketch the algorithms that transform an arbitrary ($\mathsf{GP}^2/\mathsf{C}^2$) sentence into another sentence in the normal form and recover the properties of the original input sentence from the constructed normal form sentence. We emphasize that the transformations we provide are a routine adaptation of the textbook transformation to Scott normal form: one can find similar results in [12], for the GP^2 case, and in [16] for the C^2 case.

We start with GP^2 sentences, showing that we can find an equisatisfiable formula in normal form. Note that in the normal form we allow double universal quantifiers of the form:

$$\forall x \forall y \ \ (R(x,y) \wedge x \neq y) \rightarrow \alpha(x,y)$$

which can be considered as a shorthand for the sentence:

$$\forall x \, (\#_y \, [R(x,y) \wedge x \neq y \wedge \neg \alpha(x,y)] \ = \ 0)$$

Recall that the normal form of a GP^2 sentence has the shape:

$$\varphi \ := \ \forall x \, \gamma(x) \wedge \bigwedge_{i \in [m]} \forall x \forall y \ (R_i(x,y) \wedge x \neq y) \rightarrow \alpha_i(x,y) \wedge \bigwedge_{i \in [n]} \forall x \, U_i(x) \rightarrow P_i(x),$$

where

- $\gamma(x)$ is a quantifier-free formula,
- each $\alpha_i(x,y)$ is a quantifier-free formula,
- each $P_i(x)$ is a Presburger quantified formula of the form:

$$P_i(x) \ := \ \left(\sum_{t \in [m]} \lambda_{i,t} \cdot \#_y \, [R_t(x,y) \wedge x \neq y] \ \circledast_i \ \delta_i \right).$$

The main idea is as follows. Let ψ be an arbitrary GP^2 sentence. By pushing every negation inward, we may assume that it is in negation normal form, i.e., every negation is applied only on atomic formulas. We then repeatedly replace every subformula $\nu(x)$ of one free variable x with a fresh unary predicate $U(x)$, and conjoin the original formula with the "definition" of $U(x)$:

$$\forall x \ U(x) \rightarrow \nu(x)$$

Intuitively what this means is that the subformula $\nu(x)$ is renamed as a unary predicate $U(x)$. The algorithm starts bottom-up starting from the subformula $\nu(x)$ with the lowest quantifier rank. It can be shown that the number of renaming required to obtain the normal form is linear in the length of the original sentence ψ [12, Lemma 3.3] and the whole transformation runs in linear time.

Note that unguarded unary quantification is allowed in GP^2. For universally-quantified formulas, $\forall x \ U(x)$ is already captured by the normal form. For existential-quantified $\psi := \exists x \, U(x)$, we introduce fresh unary predicates U' and binary predicate R_U, and replace ψ by

$$(\forall x \ U(x) \vee U'(x)) \ \wedge \ (\forall x \ U'(x) \rightarrow \#_y \, [R_U(x,y) \wedge U(y)] = 1) \, .$$

Note also that we can recover the model of the original sentence from the model of the constructed normal form sentence. We simply project out the fresh unary predicates introduced when constructing the normal form sentence.

Now, we turn to C^2 sentences, and recall from the body of the paper the normal form we utilized:

$$\varphi \;:=\; \forall x\, \gamma(x) \wedge \forall x\, \forall y\; (x \neq y \rightarrow \alpha(x,y)) \wedge \bigwedge_{i \in [m']} \forall x\, \exists^{=k_i} y\; (R_i(x,y) \wedge x \neq y)\,,$$

where $\gamma(x)$ and $\alpha(x,y)$ are quantifier-free formulas, R_i are binary predicates, and $k_i \in \mathbb{N}$.

The main idea of the transformation is similar to the GP^2 case: Rename every subformula $\nu(x)$ of one free variable x with a fresh unary predicate $U(x)$ and conjunct the original sentence with $\forall x\; U(x) \rightarrow \nu(x)$. It was shown in [16, 18] that there is a linear time algorithm that transforms an arbitrary C^2 sentence φ over vocabulary σ to a sentence φ^* in normal form (with additional unary predicates $V_1, \ldots, V_p$) such that:

- For every model $\mathcal{A} \models \varphi$ with domain size at least $1 + \max_{i \in [m']} k_i$, there is a model $\mathcal{A}^* \models \varphi^*$, where $\mathcal{A}^*$ is an extension of $\mathcal{A}$ to the vocabulary $\sigma^* := \sigma \cup \{V_1, \ldots, V_p\}$ with $A = A^*$ and $S^{\mathcal{A}} = S^{\mathcal{A}^*}$, for every relation symbol $S \in \sigma$.
- Conversely, for every model $\mathcal{A}^* \models \varphi^*$ with domain size at least $1 + \max_{i \in [m']} k_i$, there is a model $\mathcal{A} \models \varphi$, where $\mathcal{A}$ is obtained by projecting out the predicates $V_1, \ldots, V_p$ in $\mathcal{A}^*$.

Using these two properties, we can generalise Theorem 25 to arbitrary C^2 sentences. First, all the sizes of the models up to $1 + \max_{i \in [m']} k_i$ can be encoded inside the corresponding Presburger formula $\Psi^{\varphi}(\mathbf{x})$ mentioned in Lemma 37. Second, we can recover Π-SPEC from Π^*-SPEC, where Π and Π^* are the set of all 1-types w.r.t. σ and σ^*, via the following identity, for every 1-type π over σ:

$$x_\pi \;=\; \sum_{\pi^* \supseteq \pi \text{ and } \pi^* \text{ is 1-type over } \sigma^*} x_{\pi^*}$$

Proof of the small core lemma: lemma 32

Recall the statement of Lemma 32:

For every σ-structure $\mathcal{G}$, for every $\ell \in \mathbb{N}$, $\mathcal{G}$ has a ℓ-core $\mathcal{H}$ with size at most $(2^{2n+4m+2}) \cdot \ell$.

Proof. Recall from the body that the intuition behind the construction is to repeatedly select a 1-type or tuple that violates the ℓ-core condition and add the relevant vertices to $\mathcal{H}$. Since there are only finitely many such choices, the process eventually stabilizes, and the size of the constructed core remains bounded.

We formalize this intuition with the following $2^{2n+4m+1}$ round procedure. Let $\mathcal{H}_0$ be an empty σ-structure. For the ith round, we select either:

- a 1-type π that is realized in $\mathcal{G} \setminus \mathcal{H}_{i-1}$ by at most ℓ vertices, or
- a tuple $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps} \times \mathsf{TwoTps} \times \mathsf{OneTps}$ that is realized in $\mathcal{G} \setminus \mathcal{H}_{i-1}$ by at most ℓ pairs of vertices.

We define $\mathcal{H}_i$ as follows:

- If a 1-type π is selected, then $\mathcal{H}_i$ is obtained by extending $\mathcal{H}_{i-1}$ with all vertices from $\mathcal{G} \setminus \mathcal{H}$ with type π. Note that there are at most ℓ vertices added.

- If a tuple $\langle \pi_1, \eta, \pi_2 \rangle$ is selected, then $\mathcal{H}_i$ is obtained by extending $\mathcal{H}_{i-1}$ with all vertices from $\mathcal{G} \setminus \mathcal{H}$ involved in a pair realizing $\langle \pi_1, \eta, \pi_2 \rangle$. Note that there are at most 2ℓ vertices added.
- Otherwise if there is no such 1-type of tuple, $\mathcal{H}_i := \mathcal{H}_{i-1}$.

Finally, $\mathcal{H} := \mathcal{H}_{2^{2n+4m+1}}$.

Note that in the ith round, we add at most 2ℓ vertices to $\mathcal{H}_i$. Thus the size of $\mathcal{H}$ is bounded by $2^{2n+4m+1} \cdot 2\ell = 2^{2n+4m+2} \cdot \ell$. We claim that $\mathcal{H}$ is an ℓ-core of $\mathcal{G}$. In the ith round, if a 1-type π is selected, then for every $j \geq i$, no vertex realized π remains in $\mathcal{G} \setminus \mathcal{H}_j$. A similar property holds if a tuple is selected. Since there are 2^{n+m} possible 1-types and 2^{2n+4m} possible tuples, after $2^{2n+4m+1} \geq 2^{n+m} + 2^{2n+4m}$ rounds, no valid selection remains. Hence, $\mathcal{H}$ is an ℓ-core of $\mathcal{G}$. □

Proof of correctness of the main construction for C^2: Lemma 36

We recall details of the construction of a Presburger formula that is supposed to capture the spectrum of a C^2 formula.

Recall the definitions of various collections of 2-types: $\mathsf{TwoTps}^{\varphi,\mathsf{silent}}_{\pi_1,\pi_2} \subseteq \mathsf{TwoTps}^{\varphi}_{\pi_1,\pi_2}$ as the set of 2-types that are both forward- and backward-silent, $\mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2}$ for 2-types that are not forward-silent but are backward-silent. In the appendix we also utilize the class $\mathsf{TwoTps}^{\varphi,\triangleleft}_{\pi_1,\pi_2}$ for 2-types that are forward-silent but not backward-silent. We let $\mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2}$ denote 2-types that are neither forward-nor backward-silent.

Recall that an extended behavior, with respect to a finite set of vertices and a 1-type, consists of two components, where the first assigns elements in the set to a 2-type, and the second is a behavior vector. Recall also the construction of the formula:

- The variables are x_π for $\pi \in \mathsf{OneTps}^{\varphi}$, and $y_{\pi,\langle g,\mathbf{f}\rangle}$ for $\pi \in \mathsf{OneTps}^{\varphi}$ and $\langle g, \mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$. Intuitively, $\mathbf{x}$ is the 1-type spectrum of the finite model $\mathcal{G}$ and $y_{\pi,\langle g,\mathbf{f}\rangle}$ is the number of vertices in $\mathcal{G} \setminus \mathcal{H}$ with 1-type π and extended behavior $\langle g, \mathbf{f}\rangle$.
- The formula $\mathsf{SUM}_{\mathcal{H}}(\mathbf{x}, \mathbf{y})$ asserts that the 1-type spectrum of $\mathcal{G}$ is the sum of the 1-type spectrum of $\mathcal{H}$ and the 1-type spectrum of $\mathcal{G} \setminus \mathcal{H}$. Let s_π be the number of vertices in $\mathcal{H}$ with 1-type π.

  $$\mathsf{SUM}_{\mathcal{H}}(\mathbf{x}, \mathbf{y}) := \bigwedge_{\pi \in \mathsf{OneTps}^{\varphi}} \left(x_\pi = s_\pi + \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} y_{\pi,\langle g,\mathbf{f}\rangle} \right).$$
- The formula $\mathsf{COMP}_{\mathcal{H}}(\mathbf{y})$ guarantees that the vertices in $\mathcal{H}$ satisfies the Counting condition of Proposition 28. Let V_c be the set of vertices in $\mathcal{H}$.

  $$\mathsf{COMP}_{\mathcal{H}}(\mathbf{y}) := \bigwedge_{v \in V_c} \left(\sum_{u \in V_c \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{\pi \in \mathsf{OneTps}^{\varphi}} \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} g^{\triangleleft}(v) \cdot y_{\pi,\langle g,\mathbf{f}\rangle} = \mathbf{k}^{\varphi} \right).$$
- The formula $\mathsf{SILENT}(\mathbf{y})$ assert that 1-types realized in $\mathcal{G} \setminus \mathcal{H}$ are silent compatible.

  $$\mathsf{SILENT}_{\mathcal{H}}(\mathbf{y}) := \bigwedge_{\substack{\pi_1,\pi_2 \in \mathsf{OneTps}^{\varphi} \\ \text{are not silent compatible}}} \left(\sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} y_{\pi_1,\langle g,\mathbf{f}\rangle} = 0 \right) \vee \left(\sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_2,\mathcal{H}}} y_{\pi_2,\langle g,\mathbf{f}\rangle} = 0 \right).$$
- The formula $\mathsf{BIG}_{\mathcal{H}}(\mathbf{y})$ guarantees that 1-types and tuples realized in $\mathcal{G} \setminus \mathcal{H}$

satisfy the requirements of a core.

$$\mathsf{BIG}_{\mathcal{H}}(\mathbf{y}) := \bigwedge_{\pi \in \mathsf{OneTps}^{\varphi}} \exists z \left(z = \sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi, \mathcal{H}}} y_{\pi, \langle g, \mathbf{f} \rangle} \right) \wedge \left(z = 0 \vee z \geq (2K^{\varphi}+1)^2 \right) \wedge$$

$$\bigwedge_{\substack{\pi_1, \pi_2 \in \mathsf{OneTps}^{\varphi} \\ \eta \in \mathsf{TwoTps}^{\varphi, \triangleright \triangleleft}_{\pi_1, \pi_2}}} \exists z \left(z = \sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi_1, \mathcal{H}}} \mathbf{f}(\eta, \pi_2) \cdot y_{\pi_1, \langle g, \mathbf{f} \rangle} \right) \wedge \left(z = 0 \vee z \geq (2K^{\varphi}+1)^2 \right).$$

- The formula $\mathsf{MATCH}_{1,\mathcal{H}}(\mathbf{y})$ encodes the edge matching condition for audible 2-types and vertices in $\mathcal{G} \setminus \mathcal{H}$.

$$\mathsf{MATCH}_{1,\mathcal{H}}(\mathbf{y}) := \bigwedge_{\substack{\pi_1, \pi_2 \in \mathsf{OneTps}^{\varphi} \\ \eta \in \mathsf{TwoTps}^{\varphi, \triangleright \triangleleft}_{\pi_1, \pi_2}}} \left(\sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi_1, \mathcal{H}}} \mathbf{f}(\eta, \pi_2) \cdot y_{\pi_1, \langle g, \mathbf{f} \rangle} = \sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi_2, \mathcal{H}}} \mathbf{f}(\widetilde{\eta}, \pi_1) \cdot y_{\pi_2, \langle g, \mathbf{f} \rangle} \right).$$

- The formula $\mathsf{MATCH}_{2,\mathcal{H}}(\mathbf{y})$ encodes the edge matching condition for backward-silent but not forward-silent 2-types and vertices in $\mathcal{G} \setminus \mathcal{H}$.

$$\mathsf{MATCH}_{2,\mathcal{H}}(\mathbf{y}) := \bigwedge_{\substack{\pi_1, \pi_2 \in \mathsf{OneTps}^{\varphi} \\ \eta \in \mathsf{TwoTps}^{\varphi, \triangleright}_{\pi_1, \pi_2}}} \left(\left(\sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi_1, \mathcal{H}}} \mathbf{f}(\eta, \pi_2) \cdot y_{\pi_1, \langle g, \mathbf{f} \rangle} > 0 \right) \rightarrow \left(\sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi_2, \mathcal{H}}} y_{\pi_2, \langle g, \mathbf{f} \rangle} > 0 \right) \right).$$

  Keep in mind that we will always be considering solutions in the natural numbers. So this implication could be rewritten without summation: if for one of the extended behaviors for this triple, $\mathbf{f}(\eta, \pi_2) \cdot y_{\pi_1, \langle g, \mathbf{f} \rangle} > 0$, then one of the numbers $y_{\pi_2, \langle g, \mathbf{f} \rangle} > 0$.

Finally, $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x}) := \exists \mathbf{y}\ \mathsf{SUM}_{\mathcal{H}}(\mathbf{x}, \mathbf{y}) \wedge \mathsf{COMP}_{\mathcal{H}}(\mathbf{y}) \wedge \mathsf{SILENT}_{\mathcal{H}}(\mathbf{y}) \wedge \mathsf{BIG}_{\mathcal{H}}(\mathbf{y}) \wedge \bigwedge_{i=1,2} \mathsf{MATCH}_{i,\mathcal{H}}(\mathbf{y})$.

The correctness of this construction was stated in Lemma 36:

For every C^2 sentence φ and partial model $\mathcal{H}$ of φ, a vector $\mathbf{v}$ is a solution of $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ if and only if there exists a finite model $\mathcal{G}$ of φ such that the 1-type cardinality vector of $\mathcal{G}$ is $\mathbf{v}$, and $\mathcal{H}$ is a $(2K^{\varphi}+1)^2$-core of $\mathcal{G}$.

We now provide the proof.

Proof. **There exists a finite model $\mathcal{G}$ of φ such that the 1-type cardinality vector of $\mathcal{G}$ is $\mathbf{v}$ and $\mathcal{H}$ is a $(2K^{\varphi}+1)^2$-core of $\mathcal{G}$.** Let a_{π} be the number of vertices in $\mathcal{G}$ with 1-type π, and $b_{\pi, \langle g, \mathbf{f} \rangle}$ be the number of vertices in $\mathcal{G} \setminus \mathcal{H}$ with 1-type π and extended behavior $\langle g, \mathbf{f} \rangle$. We claim that a_{π} is a solution of $\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})$ with $b_{\pi, \langle g, \mathbf{f} \rangle}$ as the corresponding assignment to $\mathbf{y}$. Note that a_{π} is the πth component the of 1-type cardinality vector of $\mathcal{G}$, and $\sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi, \mathcal{H}}} b_{\pi, \langle g, \mathbf{f} \rangle}$ vertices in $\mathcal{G} \setminus \mathcal{H}$ with 1-type π. Let V be the set of vertices in $\mathcal{G}$ and $V_c \subseteq V$ be the set of vertices in $\mathcal{H}$.

- ($\mathsf{SUM}_{\mathcal{H}}(\mathbf{x}, \mathbf{y})$) For every π, the number of vertices with 1-type π in $\mathcal{G}$ is the sum of the number of vertices with 1-type π in $\mathcal{H}$ and $\mathcal{G} \setminus \mathcal{H}$. Recall that s_{π} is the number of vertices with 1-type π in $\mathcal{H}$. Thus it holds that

$$a_{\pi} \ = \ s_{\pi} + \sum_{\langle g, \mathbf{f} \rangle \in \mathcal{B}^{\varphi}_{\pi, \mathcal{H}}} y_{\pi, \langle g, \mathbf{f} \rangle}.$$

- ($\mathsf{COMP}_{\mathcal{H}}(\mathbf{y})$) For every vertex v in $\mathcal{H} \subseteq \mathcal{G}$, because $\mathcal{G}$ is a model of φ, v satisfies the counting condition of Proposition 28,

$$\sum_{u \in V \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) \;=\; \mathbf{k}^{\varphi}.$$

Therefore, we have

$$\begin{aligned}
&\sum_{u \in V_c \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{\pi \in \mathsf{OneTps}^{\varphi}} \sum_{f \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} g(v)^{\triangleleft} \cdot b_{\pi,f} \\
= &\sum_{u \in V_c \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{u \in V \setminus V_c} [\mathsf{bh}\,(u)]^{\triangleleft}(v) \\
= &\sum_{u \in V_c \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{u \in V \setminus V_c} \mathsf{TwoTp}^{\triangleleft}(u,v) \\
= &\sum_{u \in V \setminus \{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) \;=\; \mathbf{k}^{\varphi}.
\end{aligned}$$

- ($\mathsf{SILENT}_{\mathcal{H}}(\mathbf{y})$) Because $\mathcal{H}$ is a $(2K^{\varphi}+1)^2$-core of $\mathcal{G}$, by Lemma 33, each pair of 1-types π_1 and π_2 realized in $\mathcal{G} \setminus \mathcal{H}$ is silent compatible. Thus the constraint holds for $\mathbf{b}_{\pi,\langle g,\mathbf{f}\rangle}$.

- ($\mathsf{BIG}_{\mathcal{H}}(\mathbf{y})$) Because $\mathcal{H}$ is a $(2K^{\varphi}+1)^2$-core of $\mathcal{G}$, by the definition of a core, each 1-type π realized in $\mathcal{G} \setminus \mathcal{H}$ is realized by at least $(2K^{\varphi}+1)^2$ vertices in $\mathcal{G} \setminus \mathcal{H}$. The analogous condition holds for a tuple of 1- and 2-types. Thus the constraint holds for $\mathbf{b}_{\pi,\langle g,\mathbf{f}\rangle}$.

- ($\mathsf{MATCH}_{1,\mathcal{H}}(\mathbf{y})$) For every tuple $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps}^{\varphi} \times \mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2} \times \mathsf{OneTps}^{\varphi}$, for every pair of vertices v and u in $V \setminus V_c$ with $\mathsf{OneTp}(v) = \pi_1$ and $\mathsf{OneTp}(u) = \pi_2$, if $\mathsf{TwoTp}(v,u) = \eta$, then $\mathsf{TwoTp}(u,v) = \widetilde{\eta}$. Thus we have

$$\sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2) \cdot b_{\pi_1,\langle g,\mathbf{f}\rangle} = \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_2,\mathcal{H}}} \mathbf{f}(\widetilde{\eta},\pi_1) \cdot b_{\pi_2,\langle g,\mathbf{f}\rangle}.$$

- ($\mathsf{MATCH}_{2,\mathcal{H}}(\mathbf{y})$) For every tuple $\langle \pi_1, \eta, \pi_2 \rangle \in \mathsf{OneTps}^{\varphi} \times \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2} \times \mathsf{OneTps}^{\varphi}$, if there exists a pair of vertices v and u in $V \setminus V_c$ with $\mathsf{OneTp}(v) = \pi_1$, $\mathsf{OneTp}(u) = \pi_2$, and $\mathsf{TwoTp}(v,u) = \eta$, then

$$\sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2) \cdot b_{\pi_1,\langle g,\mathbf{f}\rangle} \;> 0$$

$$\sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_2,\mathcal{H}}} b_{\pi_2,\langle g,\mathbf{f}\rangle} \;> 0.$$

Thus the condition holds. Otherwise, if there is no such pair, then

$$\sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2) \cdot b_{\pi_1,\langle g,\mathbf{f}\rangle} \;=\; 0.$$

The condition holds trivially.

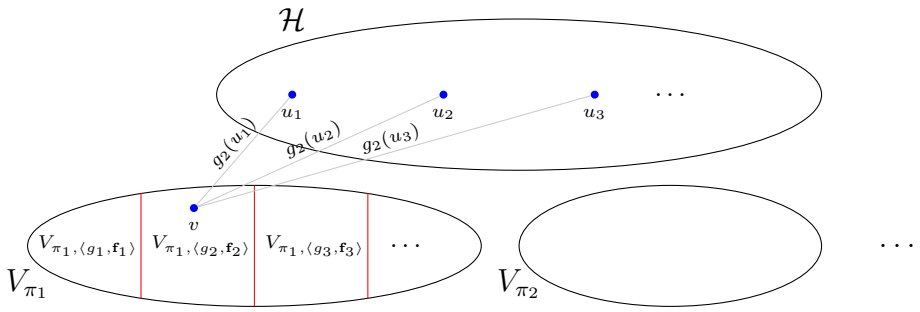


Figure 1: An illustration of the model construction extending the partial model $\mathcal{H}$ for the proof of Theorem 36.

Suppose that $\mathbf{v} \in [\![\Psi^{\varphi}_{\mathcal{H}}(\mathbf{x})]\!]$. For every $\pi \in \mathsf{OneTps}^{\varphi}$ and $\langle g, \mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$, let a_π be the πth component of $\mathbf{v}$ and $b_{\pi,\langle g,\mathbf{f}\rangle}$ be the corresponding assignment to $\mathbf{y}$. Our goal is to construct a finite model $\mathcal{G}$ of φ by extending the partial model $\mathcal{H}$. The construction is illustrated in Figure 1. We extend the partial model $\mathcal{H}$ by introducing sets of fresh vertices $V_{\pi,\langle g,\mathbf{f}\rangle}$, where the cardinality of each set is $b_{\pi,\langle g,\mathbf{f}\rangle}$. All vertices in $V_{\pi,\langle g,\mathbf{f}\rangle}$ are assigned the 1-type π. Let V_π be the union of $V_{\pi,\langle g,\mathbf{f}\rangle}$ for every $\langle g, \mathbf{f}\rangle$.

We handle edge coloring using a four-step procedure which we now outline. Firstly, we color edges between the core $\mathcal{H}$ and the relative complement of the core $\mathcal{G} \setminus \mathcal{H}$. In steps two and three, we iterate over all pairs of 1-types π_1 and π_2 realized in the relative complement of the core $\mathcal{G} \setminus \mathcal{H}$ and color edges between V_{π_1} and V_{π_2} as follows:

- In step 2, we color edges with 2-types that are neither forward-silent nor backward-silent.
- In step 3, we color edges with 2-types that are not forward-silent, but which are backward-silent.

We mention that the following invariant will be maintained during the iteration described above:

For every $\pi \in \mathsf{OneTps}^{\varphi}$ and $\langle g, \mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$, for every vertex v in $V_{\pi,\langle g,\mathbf{f}\rangle}$, for every $\pi' \in \mathsf{OneTps}^{\varphi}$ and $\eta \in \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi,\pi'} \cup \mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi,\pi'}$, the number of colored edges from v to $V_{\pi'}$ with 2-type η is either 0 or $\mathbf{f}(\eta, \pi')$.

After the iteration, in step 4 we handle the remaining edges with 2-types that are both forward- and backward-silent.

We now provide the details of the procedure.

1. For every v in $\mathcal{G} \setminus \mathcal{H}$ and u in $\mathcal{H}$, we add an edge between v and u with 2-type $g(u)$.
2. For every $\pi_1, \pi_2 \in \mathsf{OneTps}^{\varphi}$, we assign audible 2-types in $\mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2}$ and $\mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_2,\pi_1}$ between V_{π_1} and V_{π_2}. We first assume that parallel edges are allowed, Because $b_{\pi,\langle g,\mathbf{f}\rangle}$ is a solution of $\mathsf{MATCH}_{1,\mathcal{H}}(\mathbf{y})$, it is straightforward to assign (parallel) edges between V_{π_1} and V_{π_2} such that

- for every $v \in V_{\pi_1,\langle g,\mathbf{f}\rangle}$, for every $\eta \in \mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_1,\pi_2}$, the number of edges from v to V_{π_2} with 2-type η is exactly $\mathbf{f}(\eta,\pi_2)$.
- for every $v \in V_{\pi_2,\langle g,\mathbf{f}\rangle}$, for every $\eta \in \mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi_2,\pi_1}$, the number of edges from v to V_{π_1} with 2-type η is exactly $\mathbf{f}(\eta,\pi_1)$.

To eliminate parallel edges in the above construction, we again apply a swapping procedure, as in [11, 3]. Note that each vertex $v \in V_{\pi_1} \cup V_{\pi_2}$ is adjacent to at most K^φ edges. Suppose that there are parallel edges e_1 and e_2 between $v_1 \in V_{\pi_1}$ and $v_2 \in V_{\pi_2}$ with the 2-type of e_1 being η_1 and the 2-type of e_2 being η_2. Let $u_1 \in V_{\pi_1}$ and $u_2 \in V_{\pi_2}$ be vertices such that there is an edge e_3 between them with 2-type η_2, u_1 is not adjacent to v_2, and u_2 is not adjacent to v_1. Because $b_{\pi,\langle g,\mathbf{f}\rangle}$ is a solution of $\mathsf{BIG}_{\mathcal{H}}(\mathbf{y})$, there are at least $(2K^\varphi+1)^2$ edges from V_{π_1} to V_{π_2} with 2-type η_2. Since each of v_1 and v_2 has at most K^φ adjacent edges, it is always possible to find such u_1 and u_2. We then remove the edges e_2 and e_3, and replace them with fresh edges (v_1,u_2) and (v_2,u_1) with 2-type η_2. This swapping procedure preserves the number of edges connected to vertices with specific 2-types but reduces the number of parallel edges by 1, We can repeat this until all parallel edges are eliminated.

3. For every $\pi_1,\pi_2 \in \mathsf{OneTps}^\varphi$, we assign backward-silent but not forward-silent 2-types in $\mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2}$ and $\mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_2,\pi_1}$ between V_{π_1} and V_{π_2}. Again, we assume first that parallel edges are allowed, Because $b_{\pi,\langle g,\mathbf{f}\rangle}$ is a solution of $\mathsf{MATCH}_{2,\mathcal{H}}(\mathbf{y})$, if $\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^\varphi_{\pi_1,\mathcal{H}}} \mathbf{f}(\eta,\pi_2)\cdot b_{\pi_1,\langle g,\mathbf{f}\rangle} > 0$, then $\sum_{\langle g,\mathbf{f}\rangle\in\mathcal{B}^\varphi_{\pi_2,\mathcal{H}}} b_{\pi_2,\langle g,\mathbf{f}\rangle} > 0$, which implies that V_{π_2} is not empty. Thus we can assign (parallel) edges from V_{π_1} and V_{π_2} satisfying that

   - for every $v \in V_{\pi_1,\langle g,\mathbf{f}\rangle}$, for every $\eta \in \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2}$, the number of edges from v to V_{π_2} with 2-type η is exactly $\mathbf{f}(\eta,\pi_2)$.

   A similar argument applies to edges from V_{π_2} to V_{π_1}.

   - for every $v \in V_{\pi_2,\langle g,\mathbf{f}\rangle}$, for every $\eta \in \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_2,\pi_1}$, the number of edges from v to V_{π_1} with 2-type η is exactly $\mathbf{f}(\eta,\pi_1)$.

   We apply the swapping procedure to eliminate parallel edges as well. Suppose that there are parallel edges e_1 and e_2 between $v_1 \in V_{\pi_1}$ and $v_2 \in V_{\pi_2}$ with 2-types $\eta_1 \in \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi_1,\pi_2}$ for e_1 and η_2 for e_2. We consider two cases:

   - If there is a vertex $u_2 \in V_{\pi_2}$ such that there is no edges between v_1 and u_2, then we remove e_1 and add an edge between v_1 and u_2 with 2-types η_1.
   - Otherwise, suppose that there is no such u_2, which implies that for every vertex $u \in V_{\pi_2}$, there exists an edge e_3 between v_1 and u. Because there are at most $2K^\varphi(2K^\varphi+1)^2$ edges between V_{π_1} and V_{π_2}, we can find $u_1 \in V_{\pi_1}$ and $u_2 \in V_{\pi_2}$ such that no edges exists between them, and there exists an edge e_3 between v_1 and u_2 with type $\eta_3 \in \mathsf{TwoTps}^{\varphi,\triangleleft}_{\pi_1,\pi_2}$. We then remove the edges e_1 and e_3 and fresh edges (v_1,u_2) with 2-type η_1 and (v_2,u_2) with 2-type η_3.

For both cases, the swapping procedure preserves the number of edges connected to vertices with specific 2-types but reduces the number of parallel edges by 1, We can repeat it until all parallel edges are eliminated.

4. Finally, for every pair of vertices $v_1 \in V_{\pi_1}$ and $v_2 \in V_{\pi_2}$ with no edge connected, we add an edge between them with a 2-type $\eta \in \mathsf{TwoTps}^{\varphi,\mathsf{silent}}_{\pi_1,\pi_2}$.

We first claim that the 1-type cardinality vector of $\mathcal{G}$ is exactly $\mathbf{v}$. For every $\pi \in \mathsf{OneTps}^{\varphi}$, recall that the number of vertices in $\mathcal{H}$ with 1-type π is s_π. The number of vertices in $\mathcal{G} \setminus \mathcal{H}$ with 1-type π is $|V_\pi| = \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} \left|V_{\pi,\langle g,\mathbf{f}\rangle}\right| = \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} b_{\pi,\langle g,\mathbf{f}\rangle}$. Since a_π and $b_{\pi,\langle g,\mathbf{f}\rangle}$ is a solution of $\mathsf{SUM}_{\mathcal{H}}(\mathbf{x},\mathbf{y})$, we have $a_\pi = s_\pi + \sum_{f \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} b_{\pi,\langle g,\mathbf{f}\rangle}$. Recall that a_π is the ith component of $\mathbf{v}$, where i is the index of π in the ordering of $\mathsf{OneTps}^{\varphi}$. Thus the 1-type cardinality vector of $\mathcal{G}$ is exactly $\mathbf{v}$.

We next claim that $\mathcal{G}$ is a model of φ, by checking the conditions in Proposition 28. It is routine to check that all vertices and edges in $\mathcal{G}$ satisfy the 1- and 2-type conditions. Thus it is sufficient to check that the counting condition holds for every vertex v in $\mathcal{G}$.

- For every v in $\mathcal{H}$, note that

$$\sum_{u \in V_{\pi,\langle g,\mathbf{f}\rangle}} \mathsf{TwoTp}^{\triangleright}(v,u) \;=\; \sum_{u \in V_{\pi,\langle g,\mathbf{f}\rangle}} g^{\triangleleft}(v) \;=\; g^{\triangleleft}(v)\cdot\left|V_{\pi,\langle g,\mathbf{f}\rangle}\right| \;=\; g^{\triangleleft}(v)\cdot b_{\pi,\langle g,\mathbf{f}\rangle}.$$

  Because $b_{\pi,\langle g,\mathbf{f}\rangle}$ is a solution of $\mathsf{COMP}_{\mathcal{H}}(\mathbf{y})$, we have

$$\begin{aligned}
\sum_{u \in V\setminus\{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) &= \sum_{u \in V_c\setminus\{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{\pi \in \mathsf{OneTps}^{\varphi}} \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} \sum_{u \in V_{\pi,\langle g,\mathbf{f}\rangle}} \mathsf{TwoTp}^{\triangleright}(v,u) \\
&= \sum_{u \in V_c\setminus\{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{\pi \in \mathsf{OneTps}^{\varphi}} \sum_{\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}} g^{\triangleleft}(v)\cdot b_{\pi,\langle g,\mathbf{f}\rangle} \\
&= \mathbf{k}^{\varphi}
\end{aligned}$$

- For every $v \in V_{\pi,\langle g,\mathbf{f}\rangle}$, note that the extended behavior of v is $\langle g,\mathbf{f}\rangle$. Since $\langle g,\mathbf{f}\rangle \in \mathcal{B}^{\varphi}_{\pi,\mathcal{H}}$, by the compatibility condition in Definition 35, it holds that

$$\begin{aligned}
\sum_{u \in V\setminus\{v\}} \mathsf{TwoTp}^{\triangleright}(v,u) &= \sum_{u \in V_c} \mathsf{TwoTp}^{\triangleright}(v,u) + \sum_{u \in V\setminus(V_c\cup\{v\})} \mathsf{TwoTp}^{\triangleright}(v,u) \\
&= \sum_{u \in V_c} g^{\triangleright}(u) + \sum_{\pi' \in \mathsf{OneTps}^{\varphi}} \left(\sum_{\eta \in \mathsf{TwoTps}^{\varphi,\triangleright}_{\pi,\pi'}} \mathbf{f}(\eta,\pi')\cdot\eta^{\triangleright} + \sum_{\eta \in \mathsf{TwoTps}^{\varphi,\triangleright\triangleleft}_{\pi,\pi'}} \mathbf{f}(\eta,\pi')\cdot\eta^{\triangleright} \right) \\
&= \mathbf{k}^{\varphi}
\end{aligned}$$

□

11 Extensions to modulus constraints

Recall that in the body of the paper, cardinality constraints, either within GP^2 or the global cardinality constraints added to C^2, did not constrain the modulus

of a count. So, for example, in GP^2 we could not state that the number of neighbors of an element with a given property is even. And similarly, we could not add to C^2 a constraint that the total number of elements in a model satisfying a formula is an even number. We stated in the body that we can add such modulus constraints and obtain the same results.

In this appendix we will first indicate how to extend the satisfiability result for GP^2 to the case when existential quantifiers are allowed in the Presburger quantifiers, i.e., when the formula $P(x)$ is of the form:

$$P_i(x) \;:=\; \exists x_1 \cdots \exists x_n \;\sum_{t\in[m]} \lambda_t \cdot \#_y \left[R_t(x,y) \wedge x \neq y\right] \;+\; \sum_{t\in[n]} \lambda'_t \cdot x_t \;\circledast_i\; \delta.$$

Note that the existential variables captures a modulus equation, as $a \equiv b \bmod p$ can be rewritten as:

$$\exists x_1 \exists x_2 \; a + x_1 p = b + x_2 p$$

We claim that Theorem 9 still holds under such a generalization. The proof is basically the same. The only difference is that in the definition of characteristic system $\mathcal{C}^{\varphi}_{\pi}(\mathbf{x})$ defined in Definition 12 the variables $\mathbf{x}$ include all the existential variables. Similarly for the system $\mathcal{Q}^{\varphi}(\mathbf{x})$ defined in Definition 17. Then, we observe that Lemmas 16, 18, and 20 still hold for such systems and the satisfiability problem of GP^2 stays in EXP.

For the extension of C^2 with global cardinality and modulus constraints, the situation is more obvious. We have already shown how to get a (modulus-constraint free) semilinear representation for the type spectrum vectors. We can compose this with modulus constraints in addition to linear arithmetic constraints and solve using standard techniques for Presburger arithmetic. Rewriting modulus constraint into a single linear equation does not change the number of constraints in each disjunct in the formula $\Psi^{\varphi}(\mathbf{x})$. We emphasize again that the argument here is completely generic, and the same comment holds to show that for *any* decidability extension of Presburger arithmetic (e.g. Büchi Arithemetic or Semenov Arithmetic [21]), C^2 with global cardinality constraints of that form is decidable.

Expressiveness separation of the considered languages

We mentioned that the languages we consider in this paper – GP^2 and C^2 with global cardinality constraints – are incomparable in expressiveness.

GP^2 contains sentences not in C^2 with constraints, because (as we showed in the paper) the spectrum of GP2 may not be semilinear, while for C^2 with constraints the specctrum is always semilinear.

On the other hand, even ordinary C^2 contains sentences not in GP^2: just consider the sentence restricting the cardinality of the model to some fixed number, like 2. In GP^2 every sentence that is consistent has models of unbounded finite cardinality, since we can always clone a connected component without disturbing the truth value of any sentence.

References

[1] Franz Baader. Expressive cardinality restrictions on concepts in a description logic with expressive number restrictions. *SIGAPP Appl. Comput. Rev.*, 19(3), 2019.

[2] Bartosz Bednarczyk, Maja Orlowska, Anna Pacanowska, and Tony Tan. On classical decidable logics extended with percentage quantifiers and arithmetics. In *FSTTCS*, 2021.

[3] Michael Benedikt, Egor Kostylev, and Tony Tan. Two Variable Logic with Ultimately Periodic Counting. *SIAM Journal on Computing*, 53(4):884–968, 2024.

[4] Michael Benedikt, Chia-Hsuan Liu, Boris Motik, and Tony Tan. Verification of graph neural networks via logical characterizations. In *ICALP*, 2024.

[5] Dmitry Chistikov and Christoph Haase. The Taming of the Semi-Linear Set. In *ICALP*, 2016.

[6] F. Eisenbrand and G. Shmonin. Carathéodory bounds for integer cones. *Oper. Res. Lett.*, 34(5):564–568, 2006.

[7] D. R. Ford and D. R. Fulkerson. *Flows in Networks*. Princeton University Press, USA, 2010.

[8] Christoph Haase and Georg Zetzsche. Presburger arithmetic with stars, rational subsets of graph groups, and nested zero tests. In *LICS*, 2019.

[9] Narendra Karmarkar. A new polynomial-time algorithm for linear programming. In *STOC*, 1984.

[10] Yevgeny Kazakov. A polynomial translation from the two-variable guarded fragment with number restrictions to the guarded fragment. In *JELIA*, 2004.

[11] Eryk Kopczynski and Tony Tan. Regular graphs and the spectra of two-variable logic with counting. *SIAM J. Comput.*, 44(3):786–818, 2015.

[12] Chia-Hsuan Lu and Tony Tan. On two-variable guarded fragment logic with expressive local Presburger constraints. *Logical Methods in Computer Science*, 20, 2024.

[13] C. Papadimitriou. On the complexity of integer programming. *J. ACM*, 28(4):765–768, 1981.

[14] Rohit Parikh. On Context-Free Languages. *JACM*, 13(4):570–581, 1966.

[15] Loic Pottier. Minimal solutions of linear diophantine systems: Bounds and algorithms. In *RTA*, 1991.

[16] Ian Pratt-Hartmann. Complexity of the two-variable fragment with counting quantifiers. *Journal of Logic Language and Information*, 14:369–395, 2005.

[17] Ian Pratt-Hartmann. Complexity of the guarded two-variable fragment with counting quantifiers. *Journal of Logic and Computation*, 17(1):133–155, 2007.

[18] Ian Pratt-Hartmann. The two-variable fragment with counting. In *WoLLIC*, 2010.

[19] "Johann" Sebastian Rudolph. Presburger concept cardinality constraints in very expressive description logics - allegro sexagenarioso ma non ritardando. In *Description Logic, Theory Combination, and All That - Essays Dedicated to Franz Baader on the Occasion of His 60th Birthday*, 2019.

[20] Helmut Seidl, Thomas Schwentick, Anca Muscholl, and Peter Habermehl. Counting in trees for free. In *ICALP*, 2004.

[21] Aleksei L. Semenov. Logical theories of one-place functions on the set of natural numbers. *Math. USSR Izv.*, 22(3):587–618, 1984.

[22] Boris Trakhtenbrot. The Impossibility of an Algorithm for the Decidability Problem on Finite Classes. *Proceedings of the USSR Academy of Sciences*, 70(4):569–572, 1950.